

UC10 — Claims & Authorization

Data-Quality Anomaly Monitor

Data Cleaning Pipeline — Before vs After Visualization Report

Raw: 10,000 rows × 50 cols → Cleaned: 10,000 rows × 86 cols
0 records deleted | 36 new DQ/anomaly flag columns added | 6,626 records flagged with ≥1 DQ issue

14 Visualization Sections:

Overview | Missing Values | Latency | Financials | Dates | Auth | SLA/Pipeline
Provider/Bene | Record-Type Missingness | DQ Score | All Flags | Cleaning Table | Correlation | Schema

UC10 – Data Cleaning Pipeline: Executive Overview

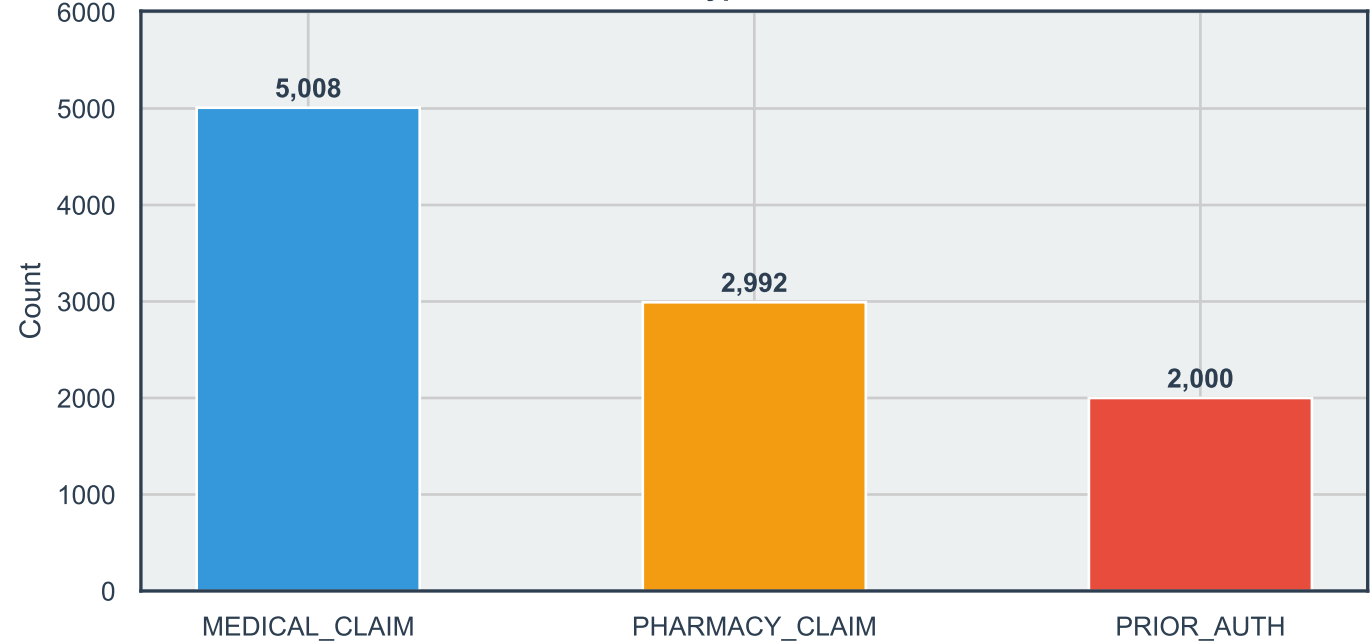
unchanged

+36 new flag cols

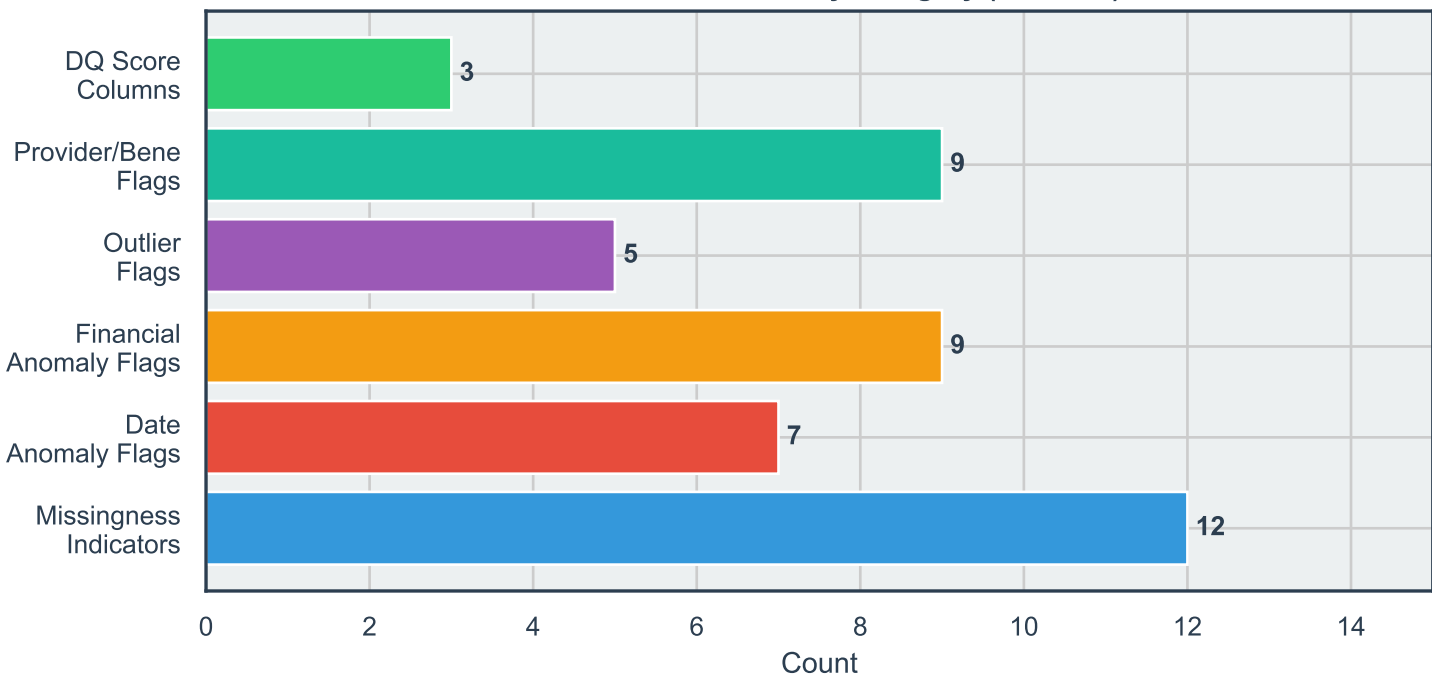
identifiers preserved

anomalies flagged, not removed

Record Type Distribution



New Columns Added by Category (Total: 36)

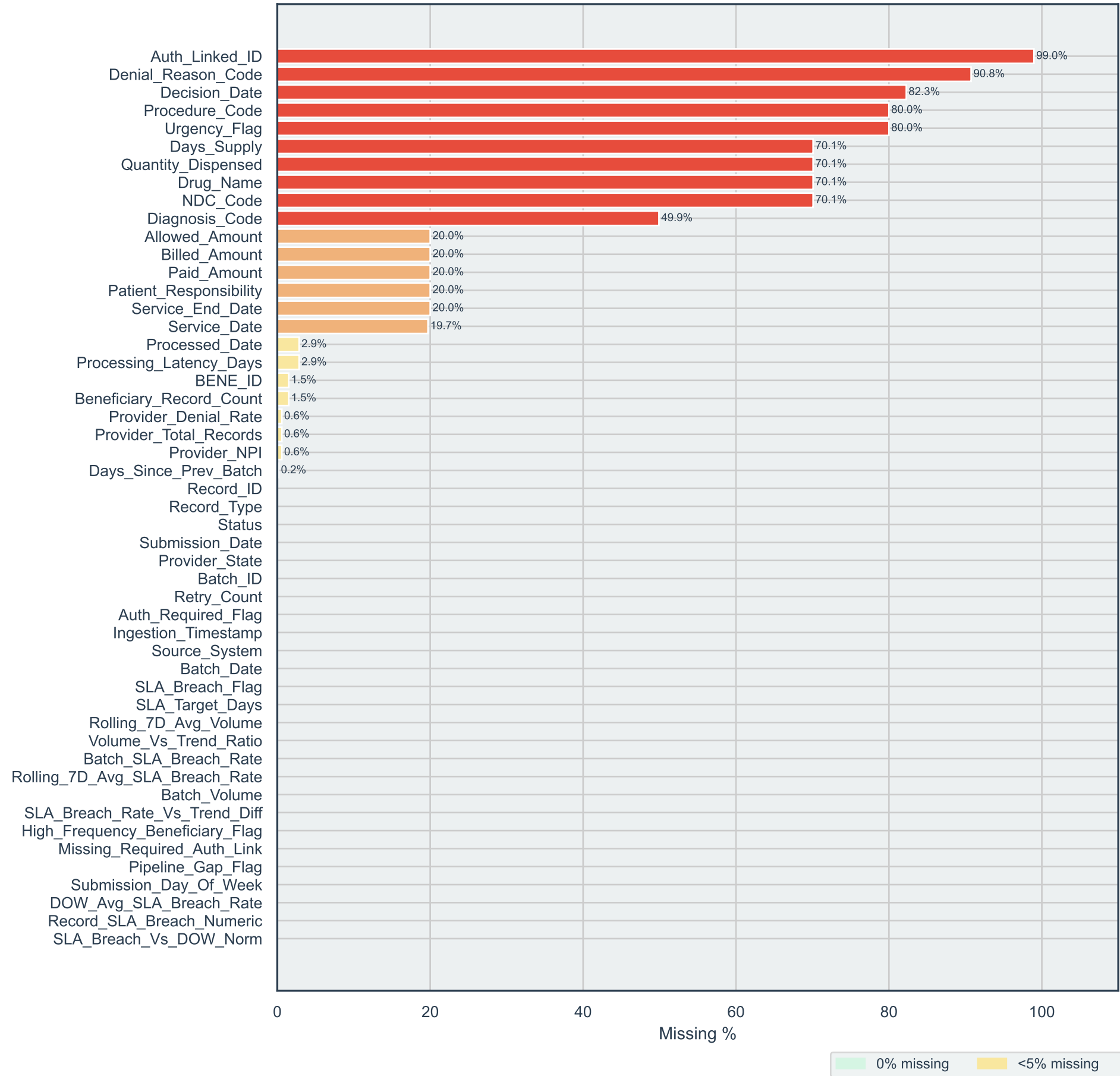


SECTION 1 — Missing Value Analysis

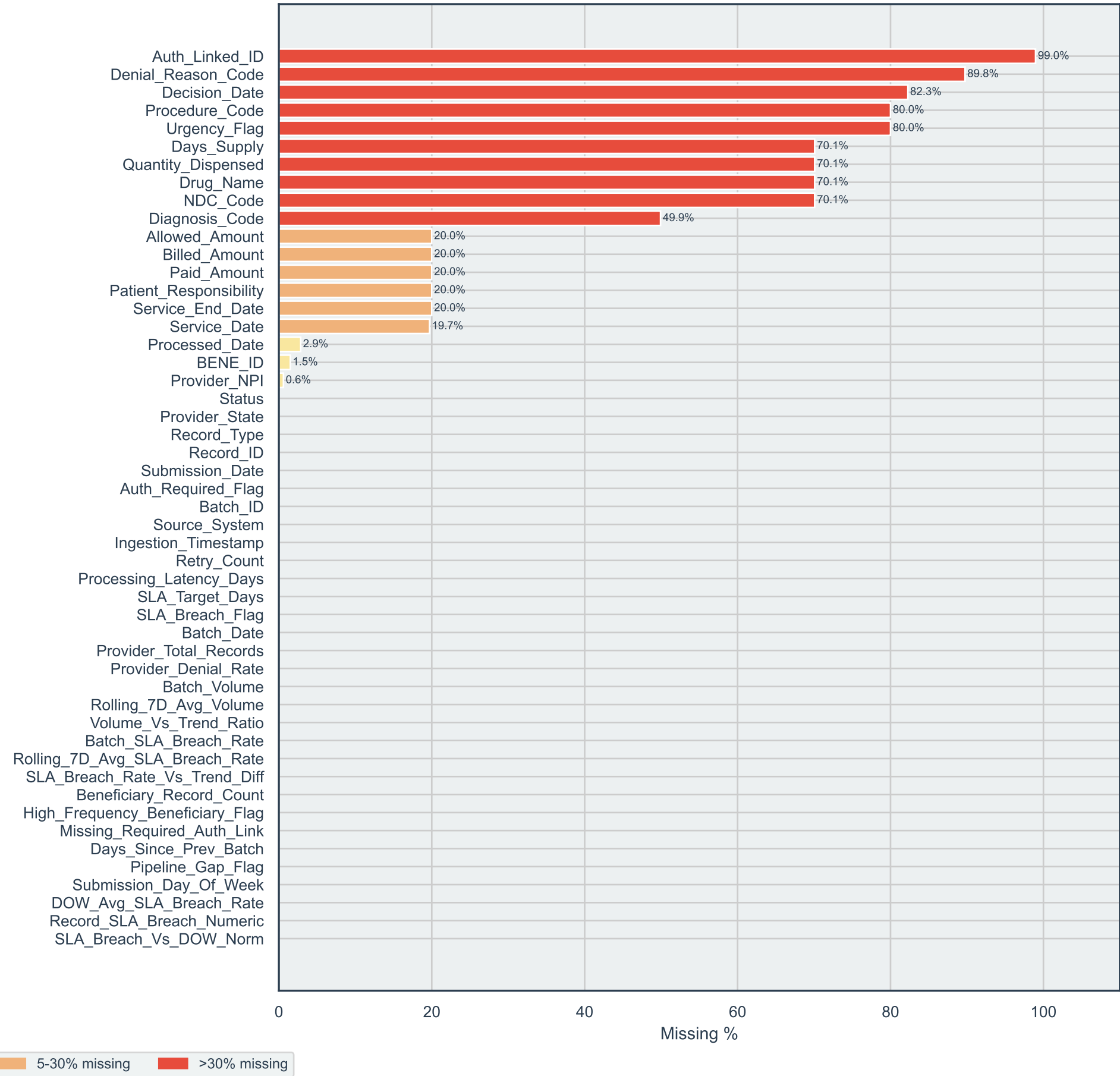
Before vs After | Identifiers never imputed | Missingness indicators created

Missing Values — Before vs After Cleaning

BEFORE Cleaning (Raw)



AFTER Cleaning (Original Columns Only)

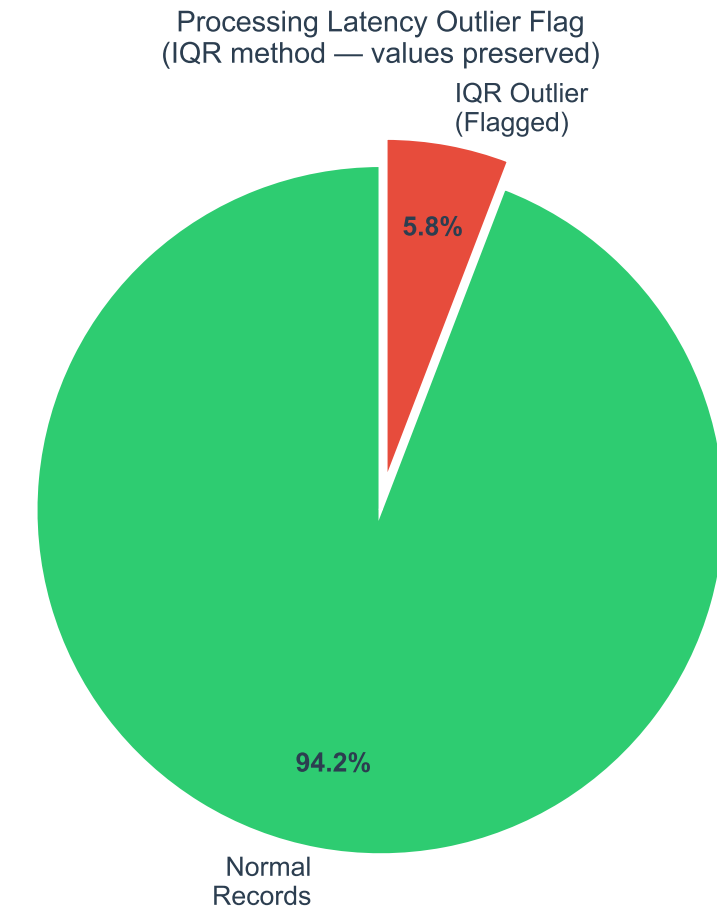
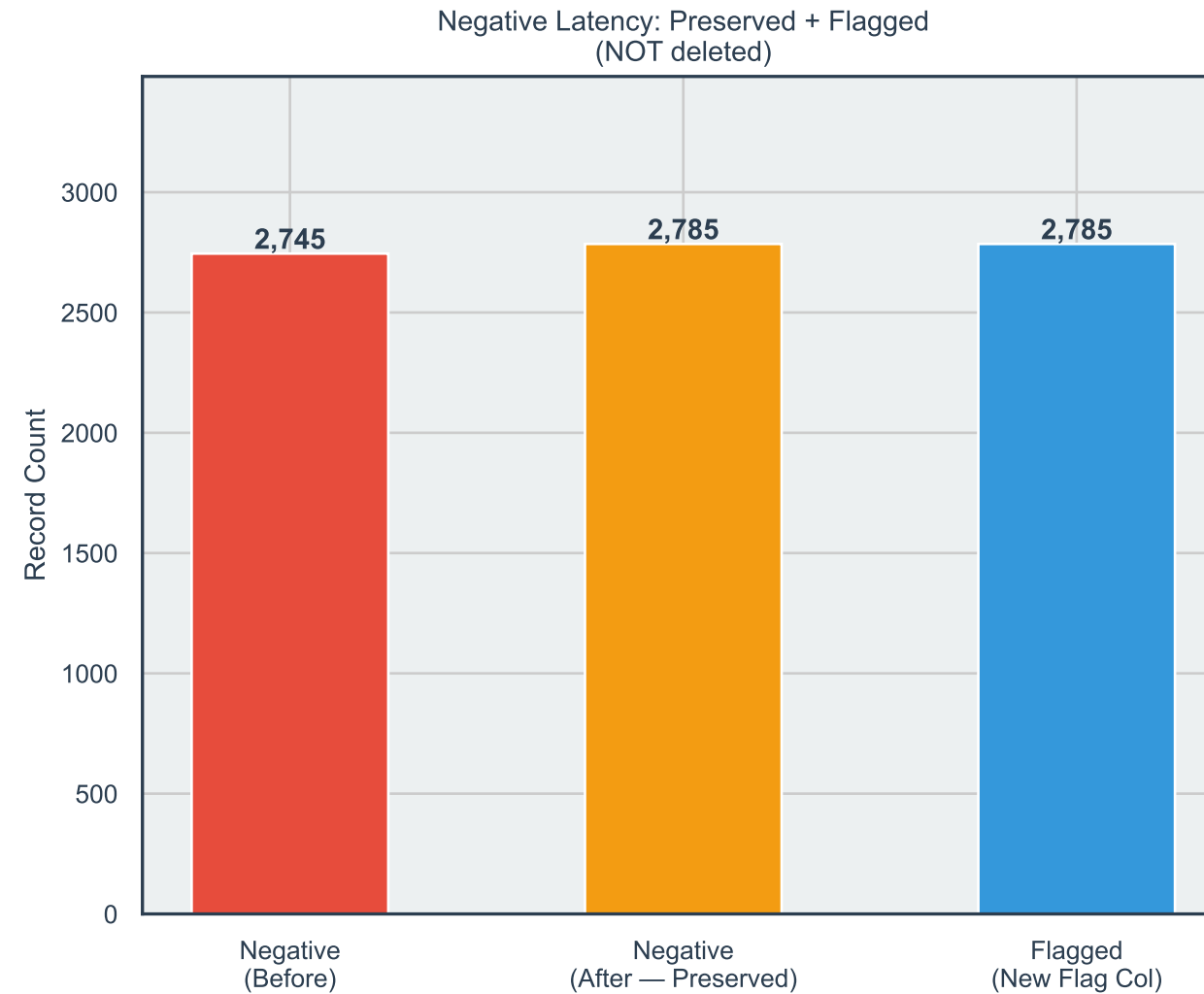
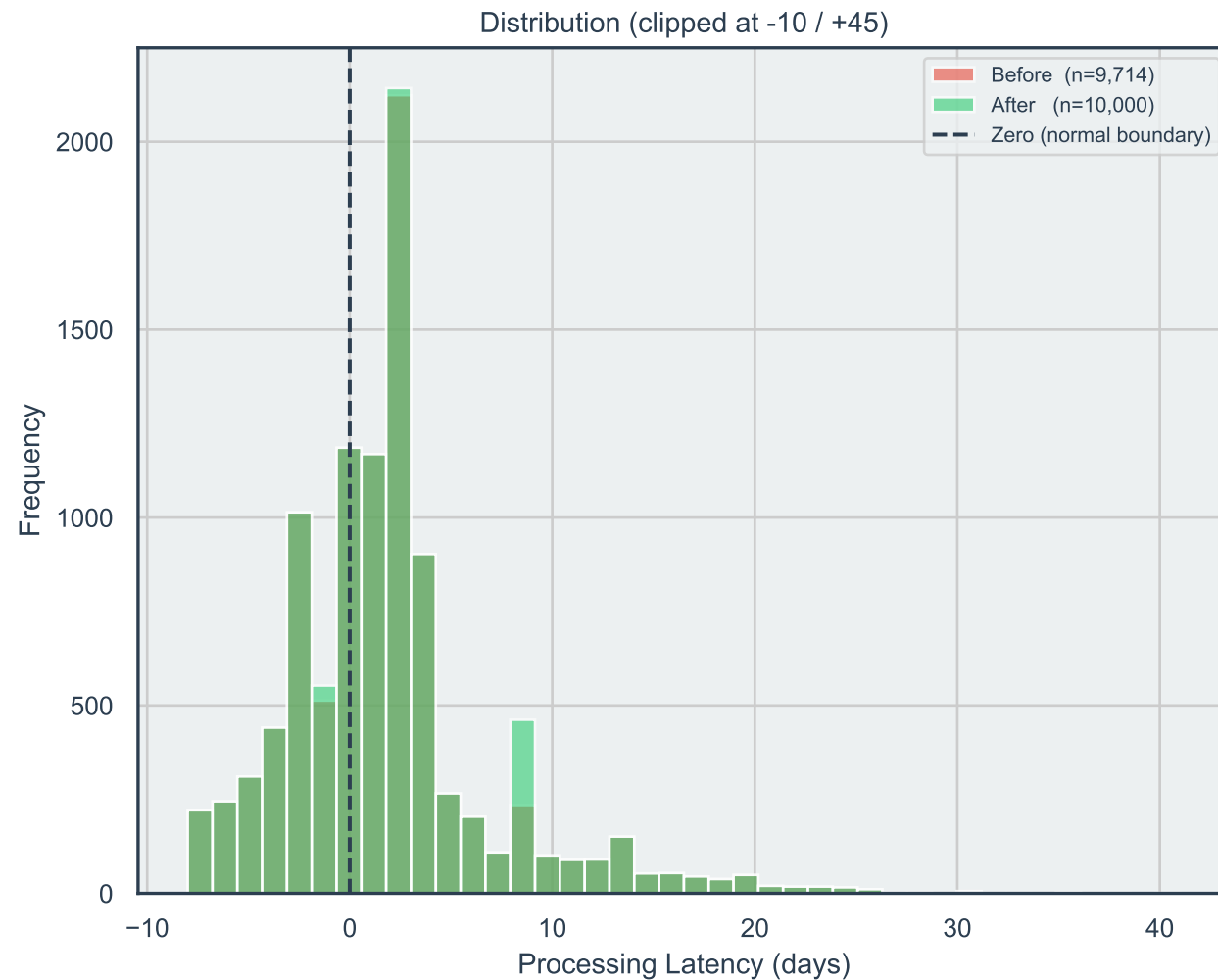


After cleaning: identifiers (BENE_ID, Provider_NPI) remain NaN — values are NOT fabricated. Missingness indicators created instead.

SECTION 2 — Processing Latency

2,785 negative-latency records PRESERVED with flag | IQR outliers flagged, not removed

Processing Latency Analysis — Before vs After

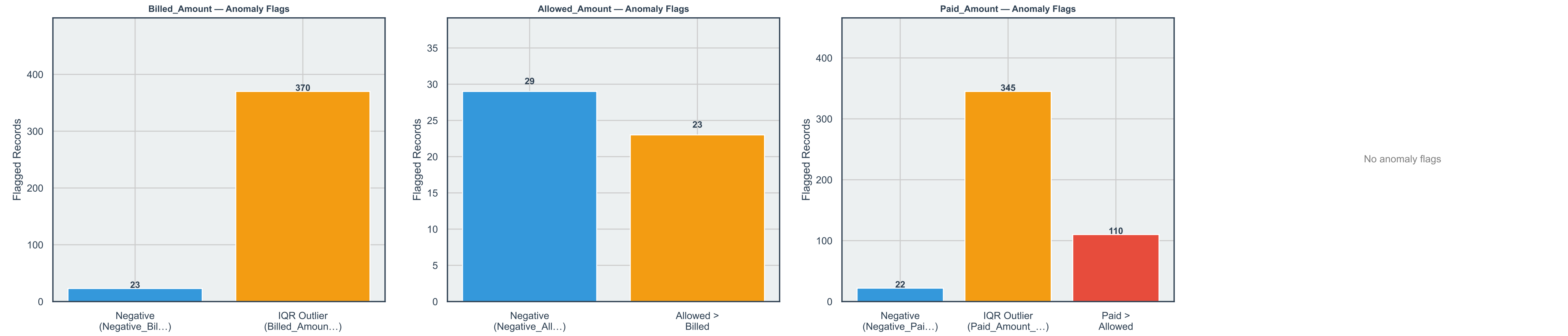
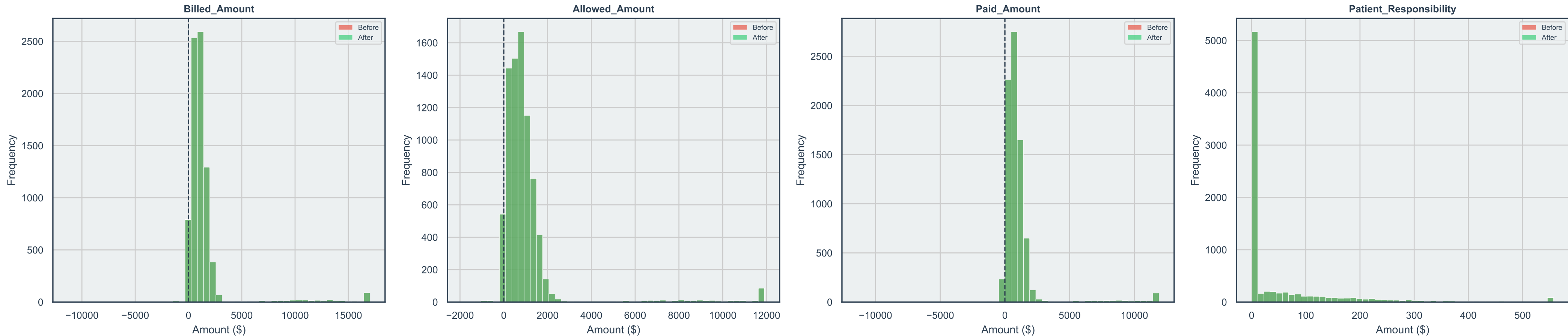


Key change: 2,785 negative-latency records PRESERVED with Negative_Processing_Latency_Flag=1 | 583 IQR outliers FLAGGED, not removed

SECTION 3 — Financial Amounts

Negative amounts preserved | Paid>Allowed flagged | IQR outliers flagged

Financial Amounts — Distribution Before vs After (Anomalies Preserved)

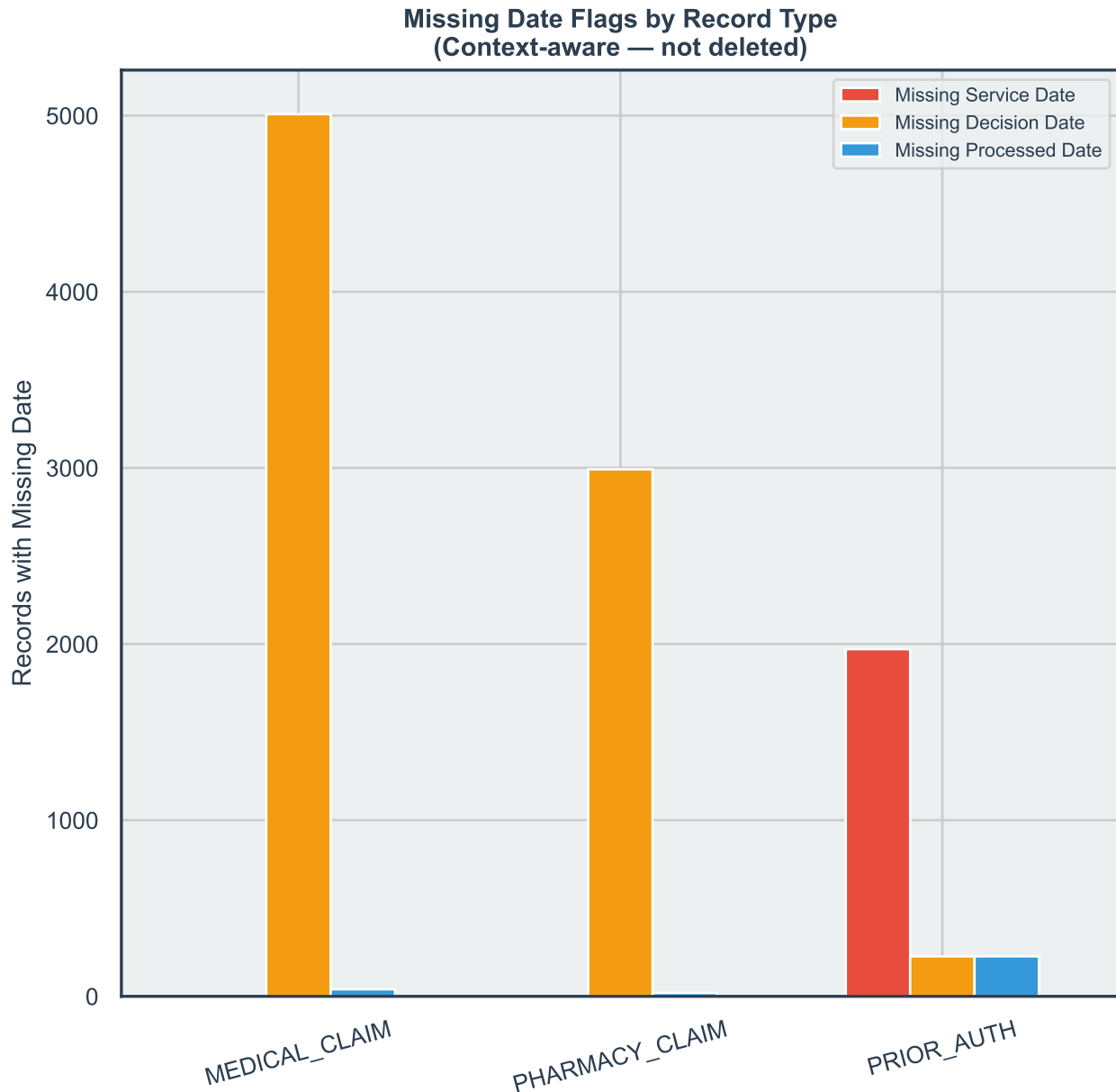
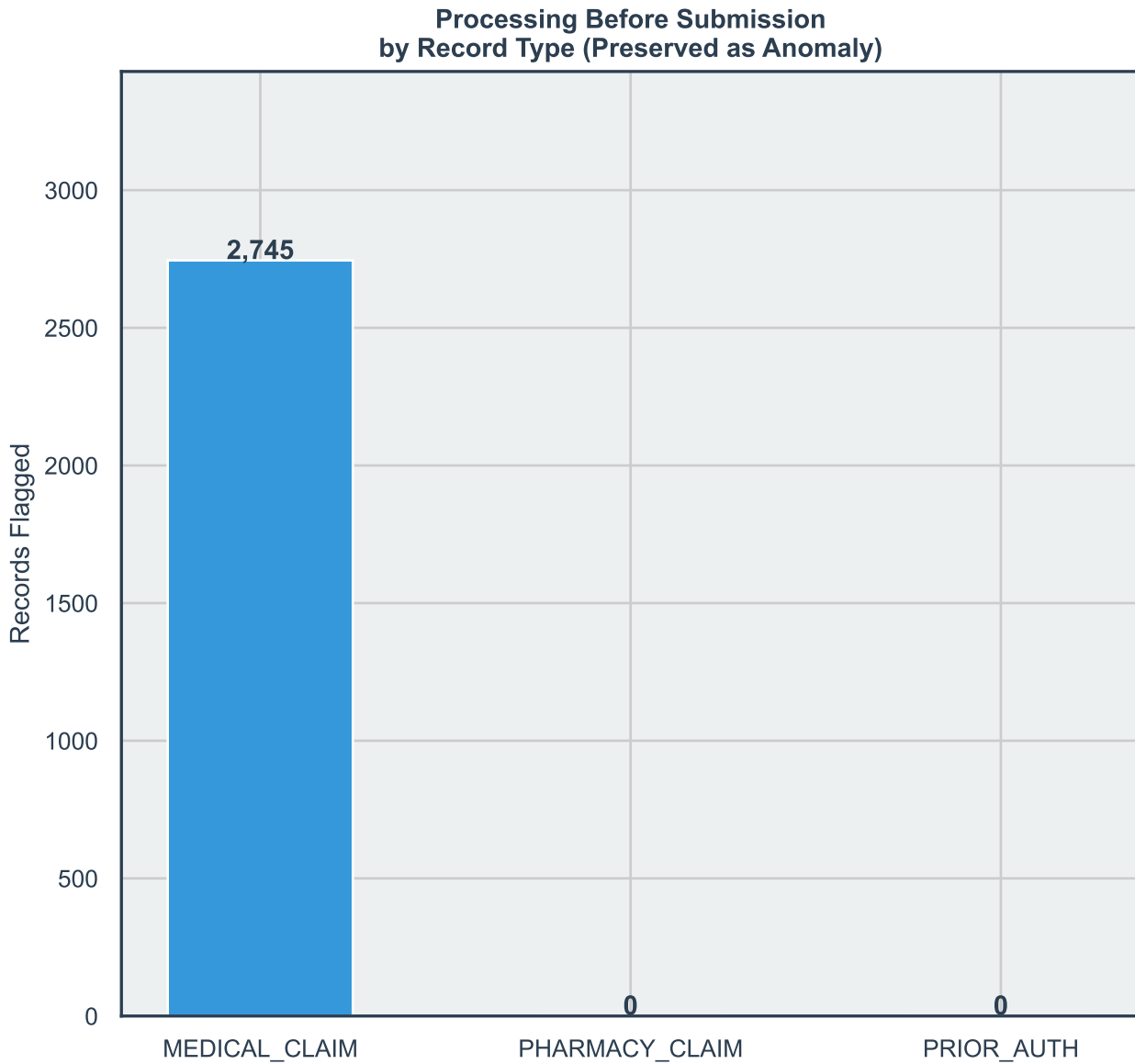
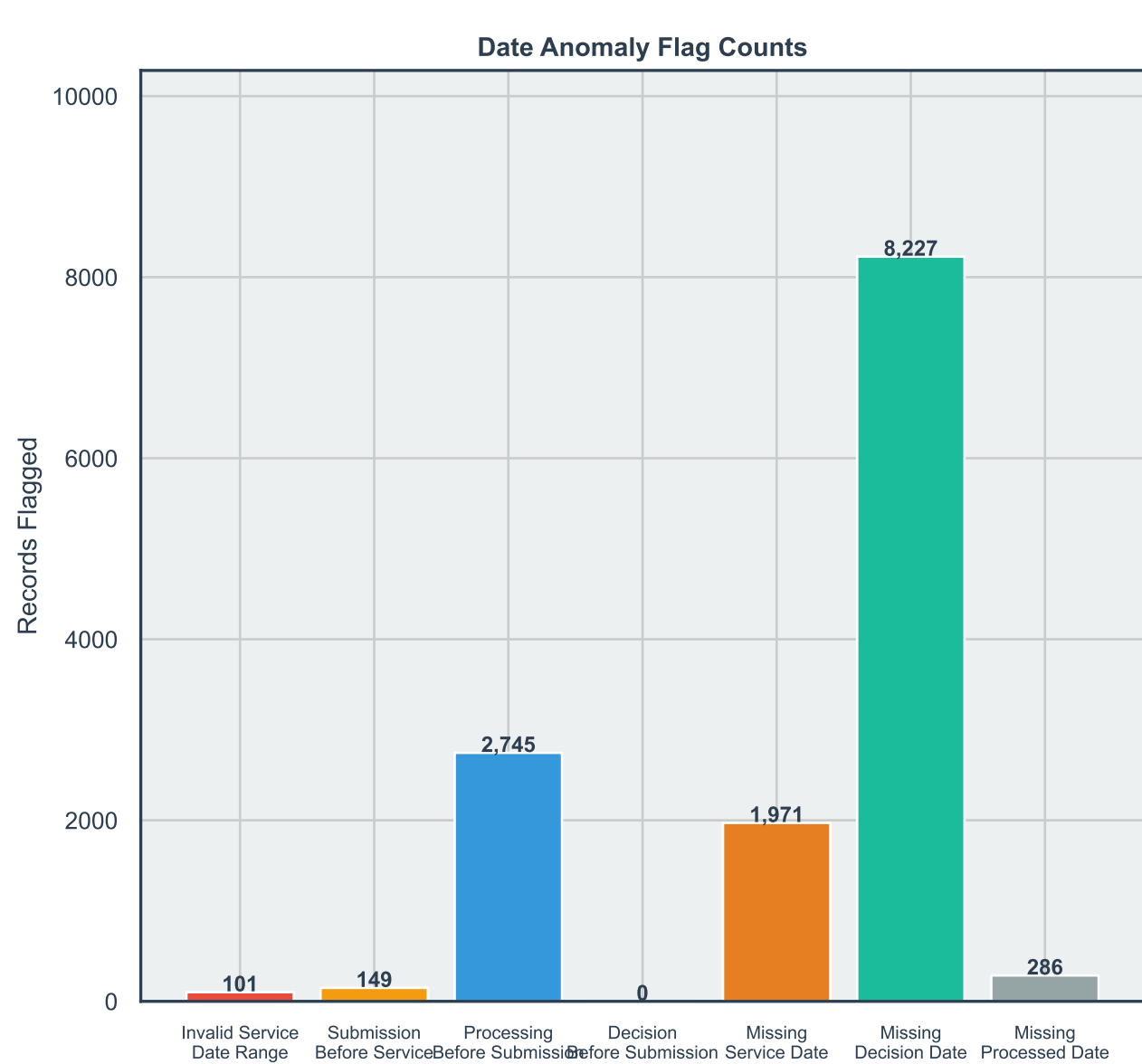


Key: Negative amounts and outliers are PRESERVED with flag columns. No blanket deletion or clipping applied.

SECTION 4 — Date Anomalies

4 date-order flags | 2,845+ date anomaly records preserved, not deleted

Date Anomalies — Flagged After Cleaning (Not Removed)

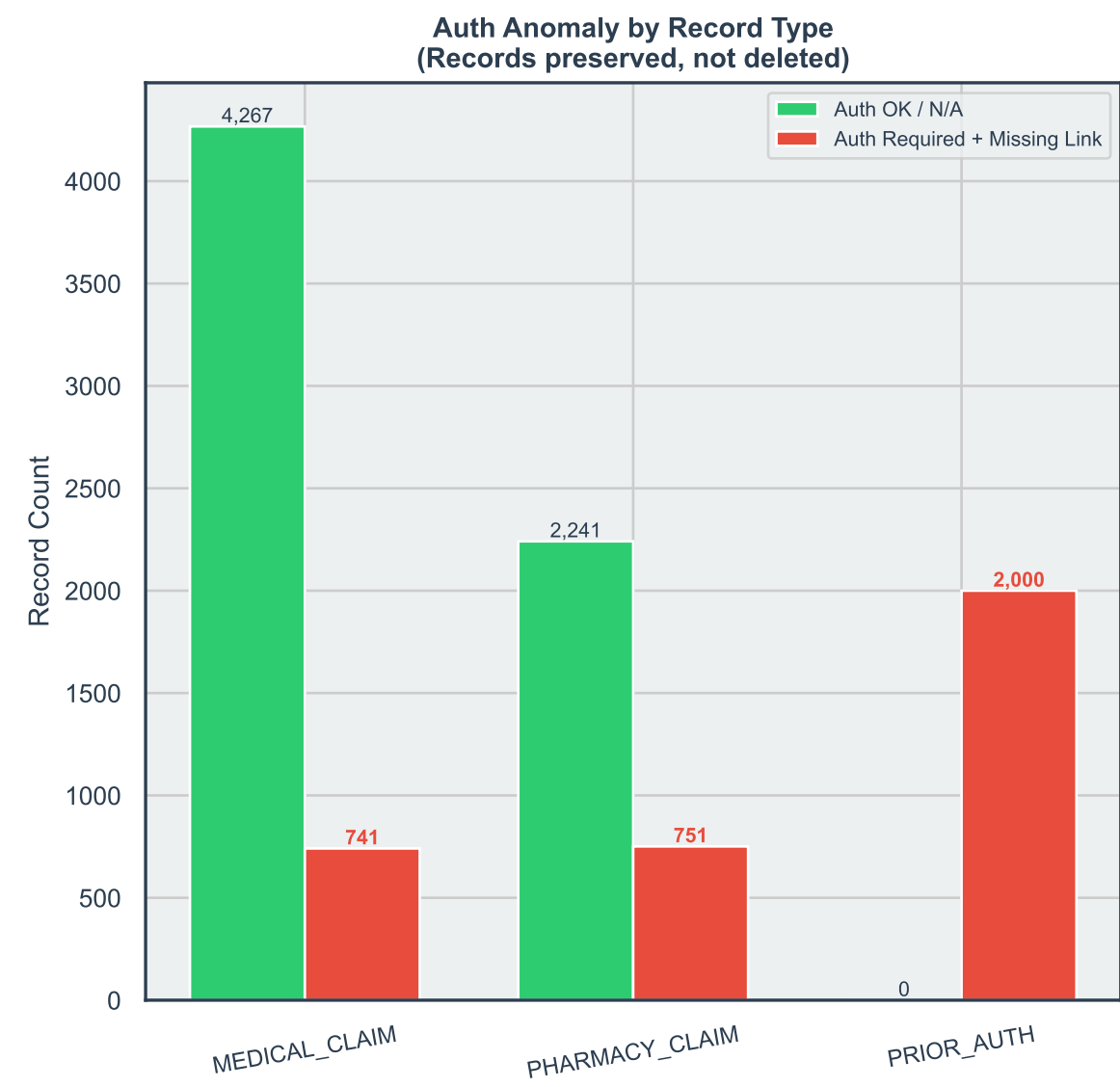
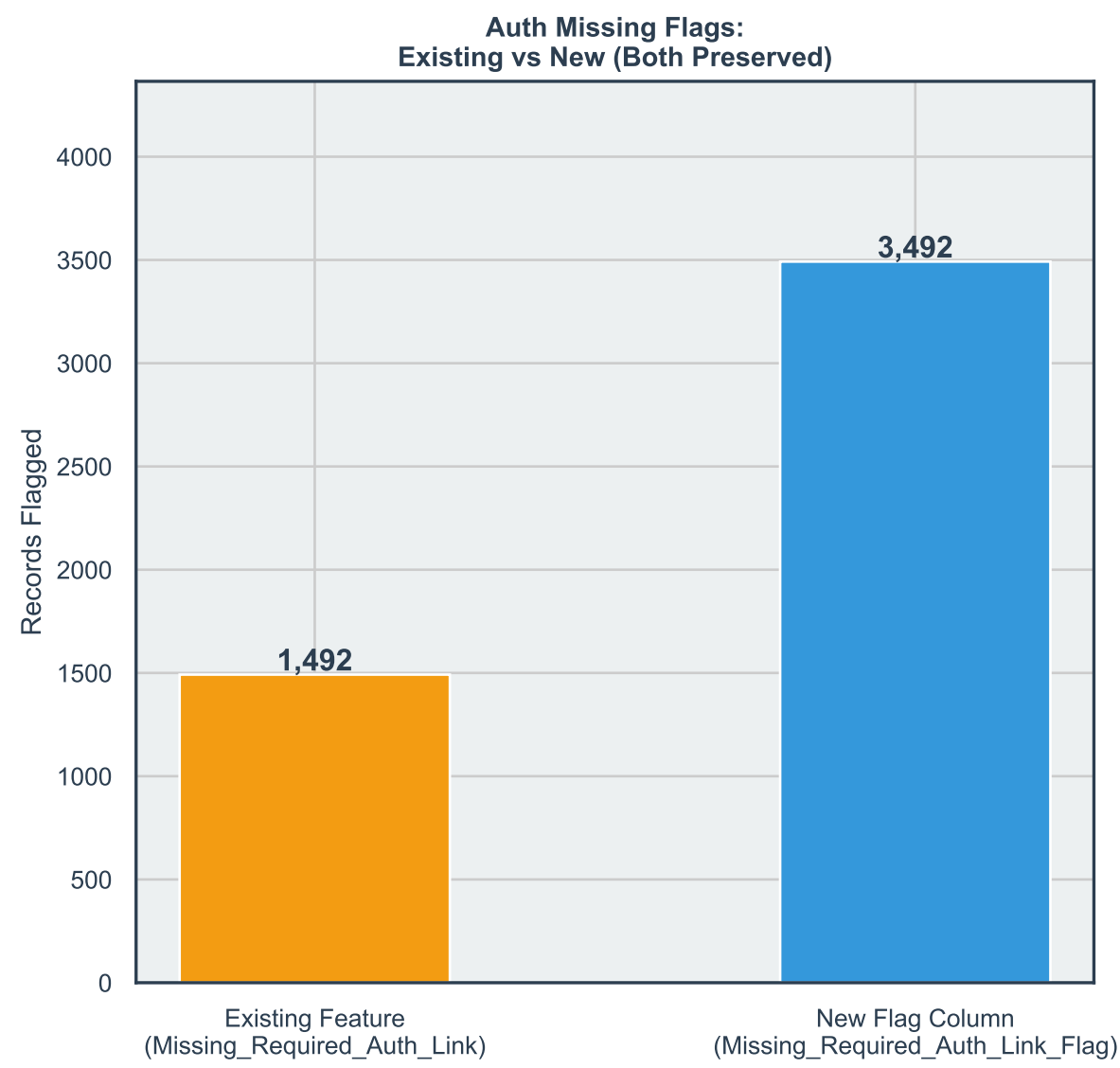
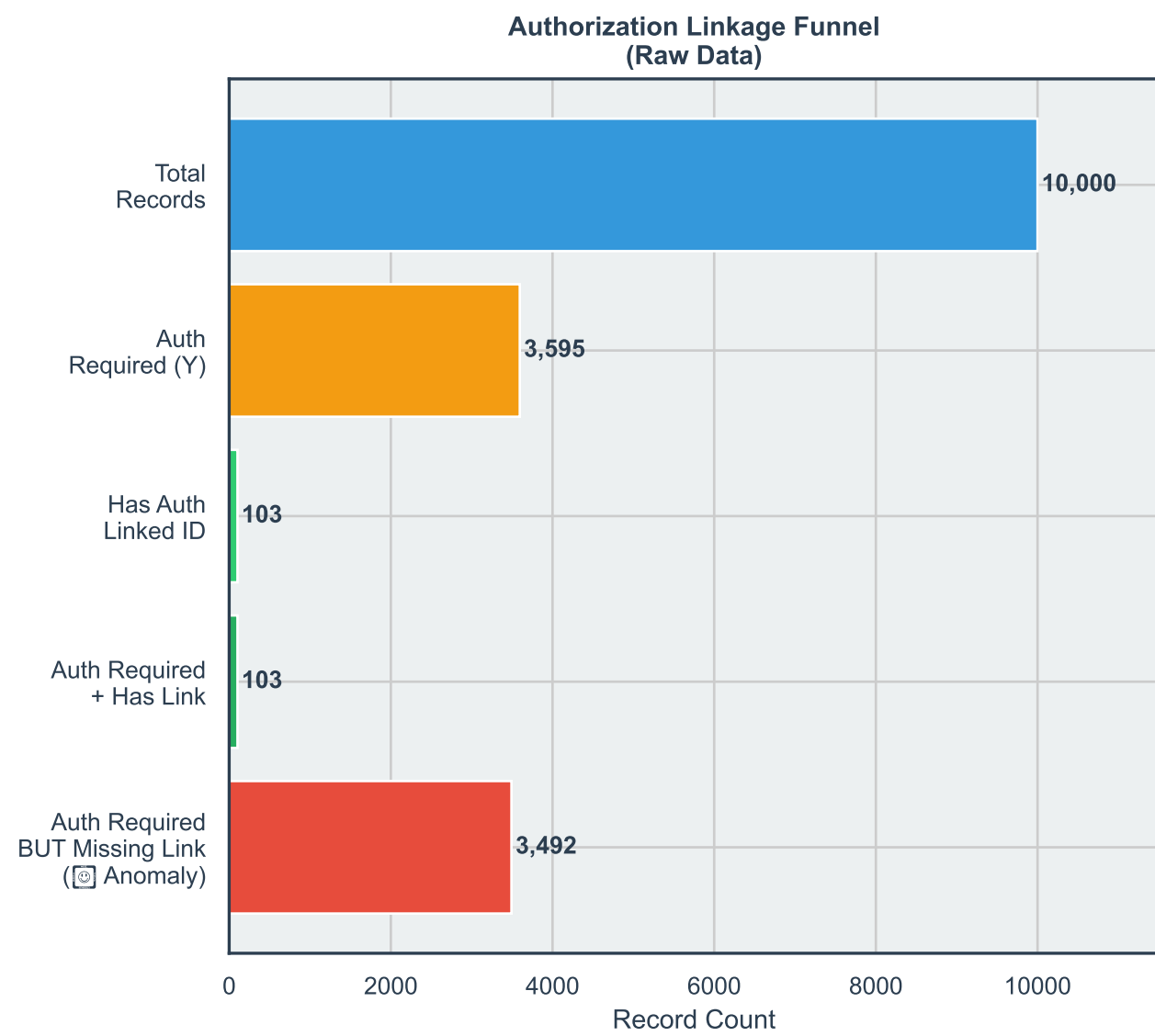


Processing_Before_Submission (2,745 records) is a genuine pipeline clock anomaly. All records retained with flag=1.

SECTION 5 — Authorization Anomalies

3,492 Auth-required but missing link records PRESERVED with flag

Authorization Anomalies — Before vs After

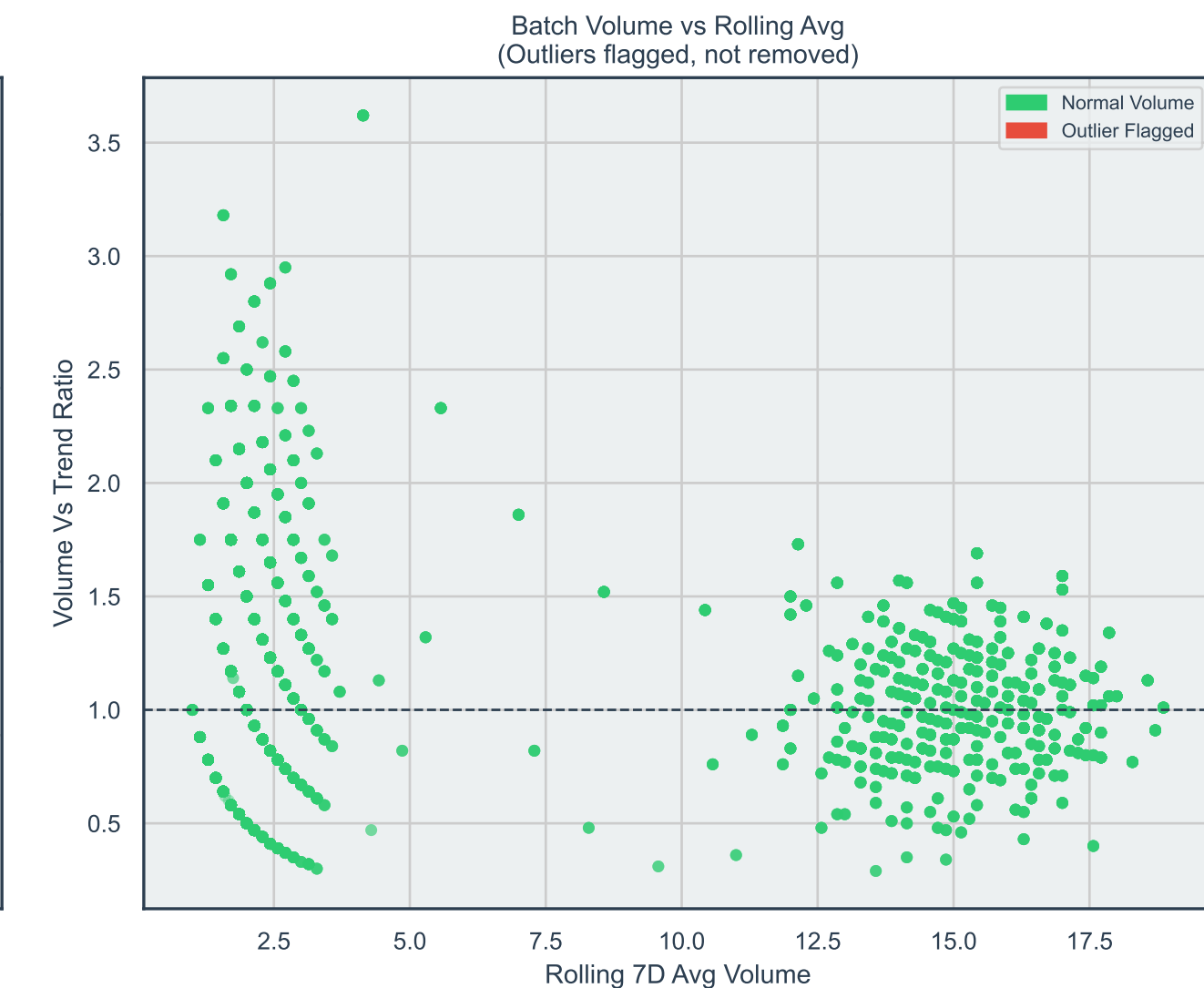
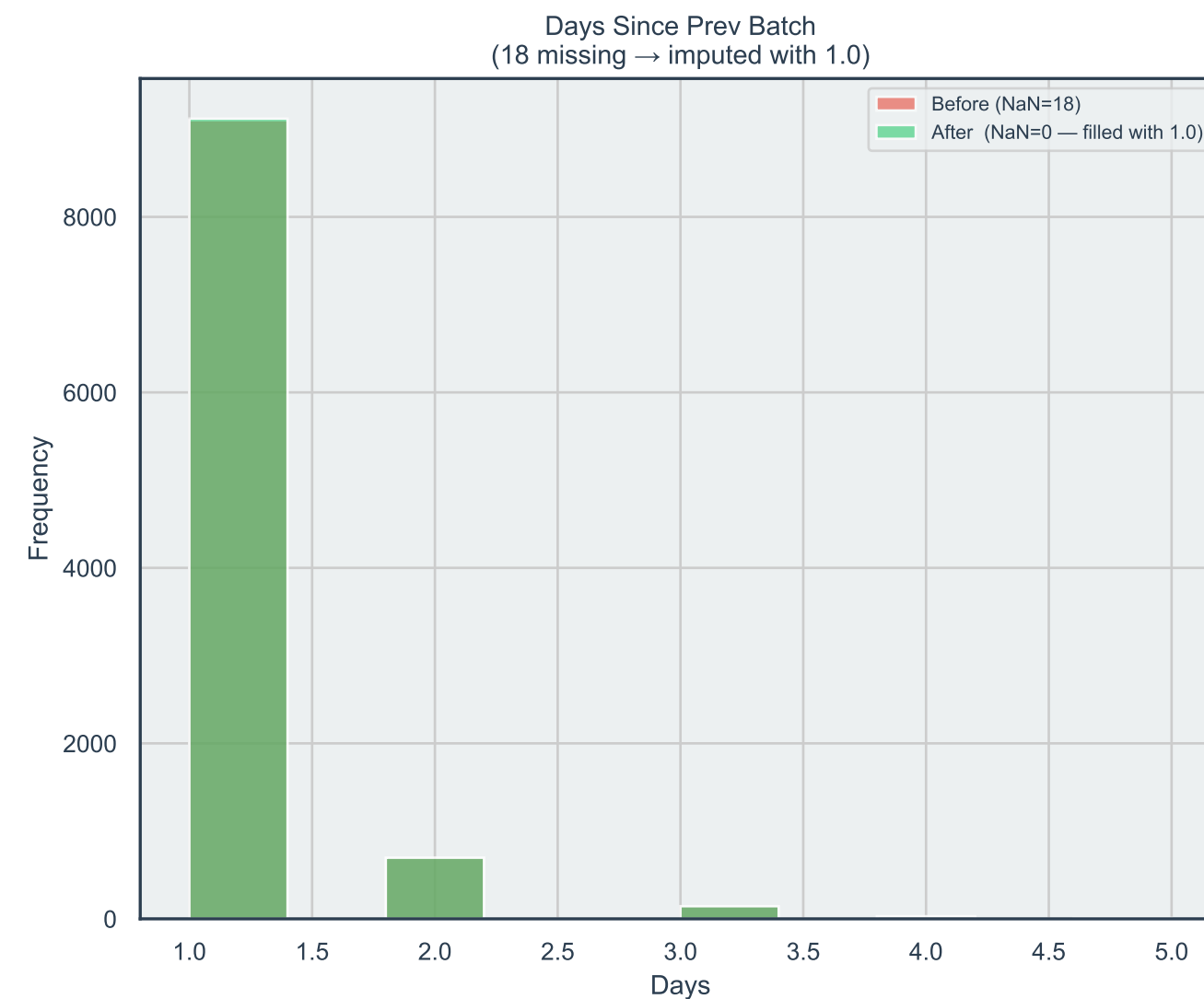
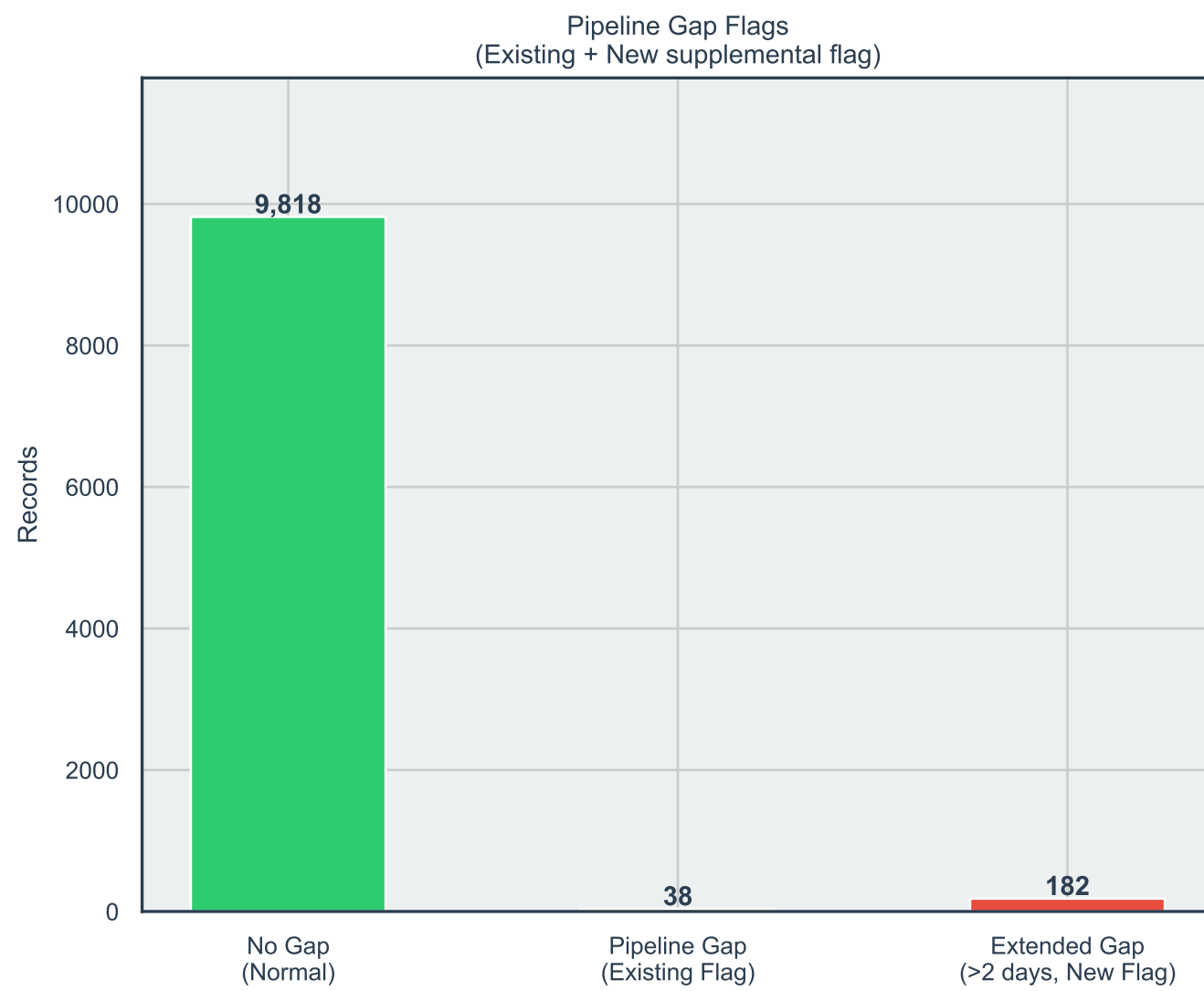
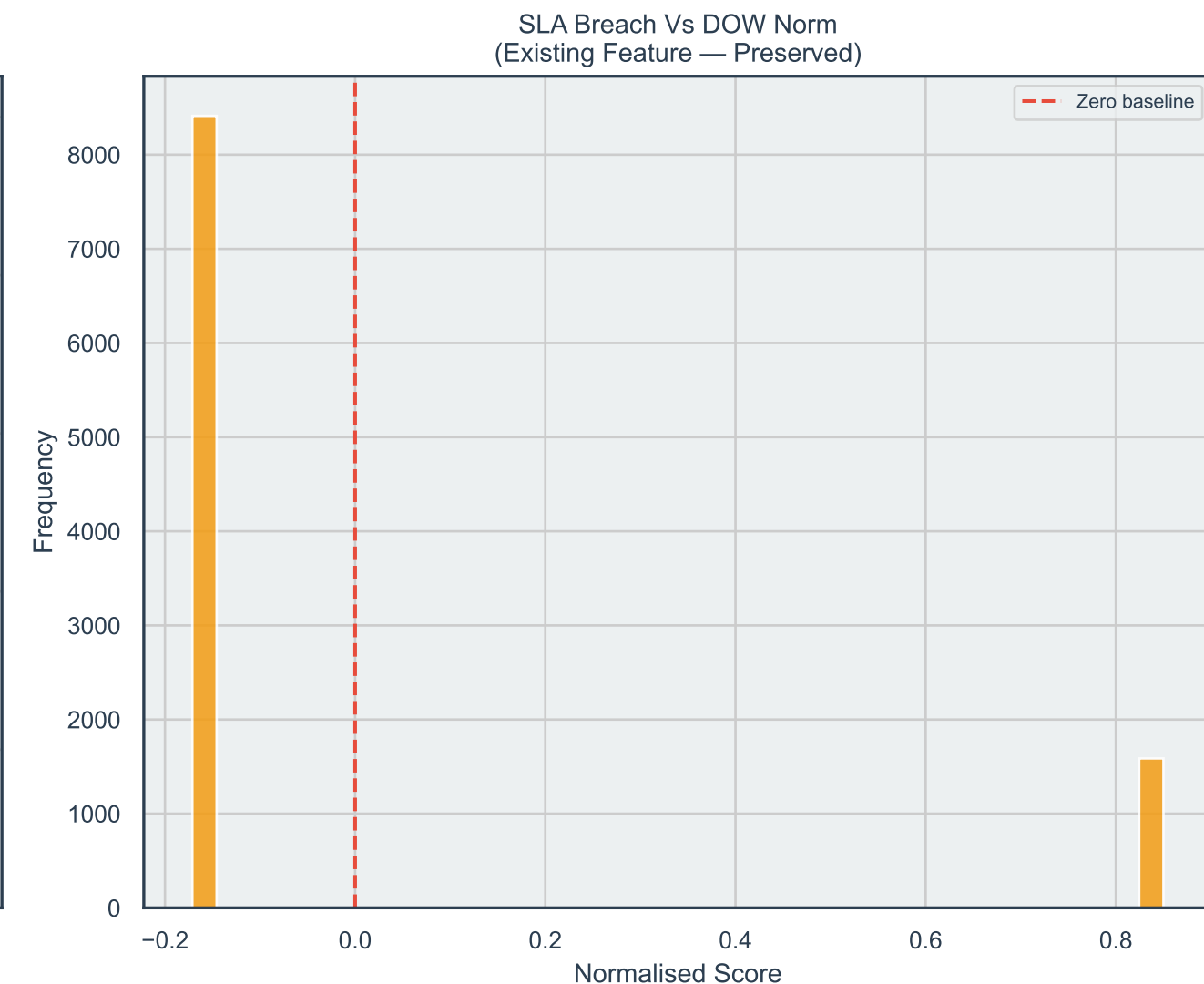
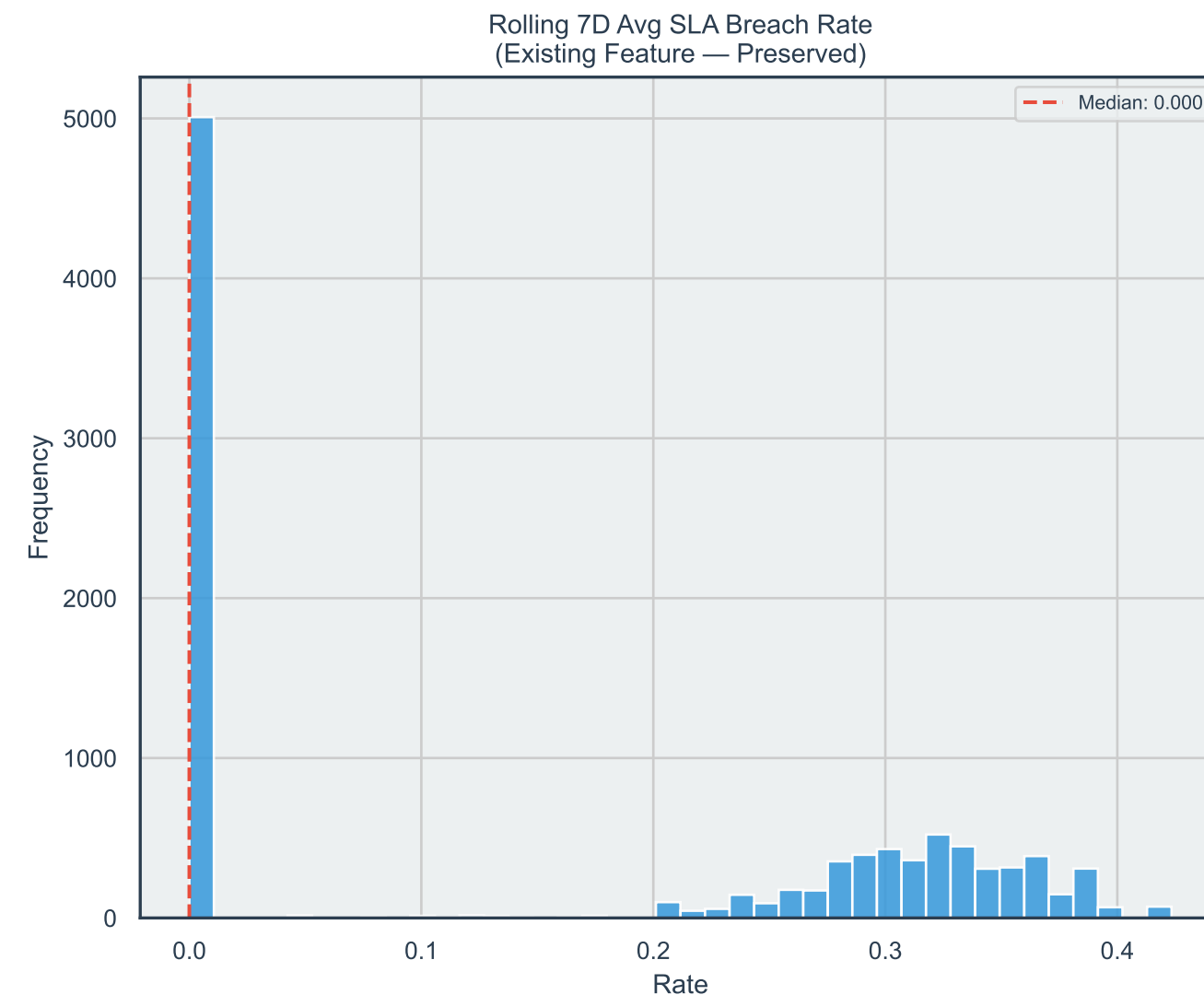
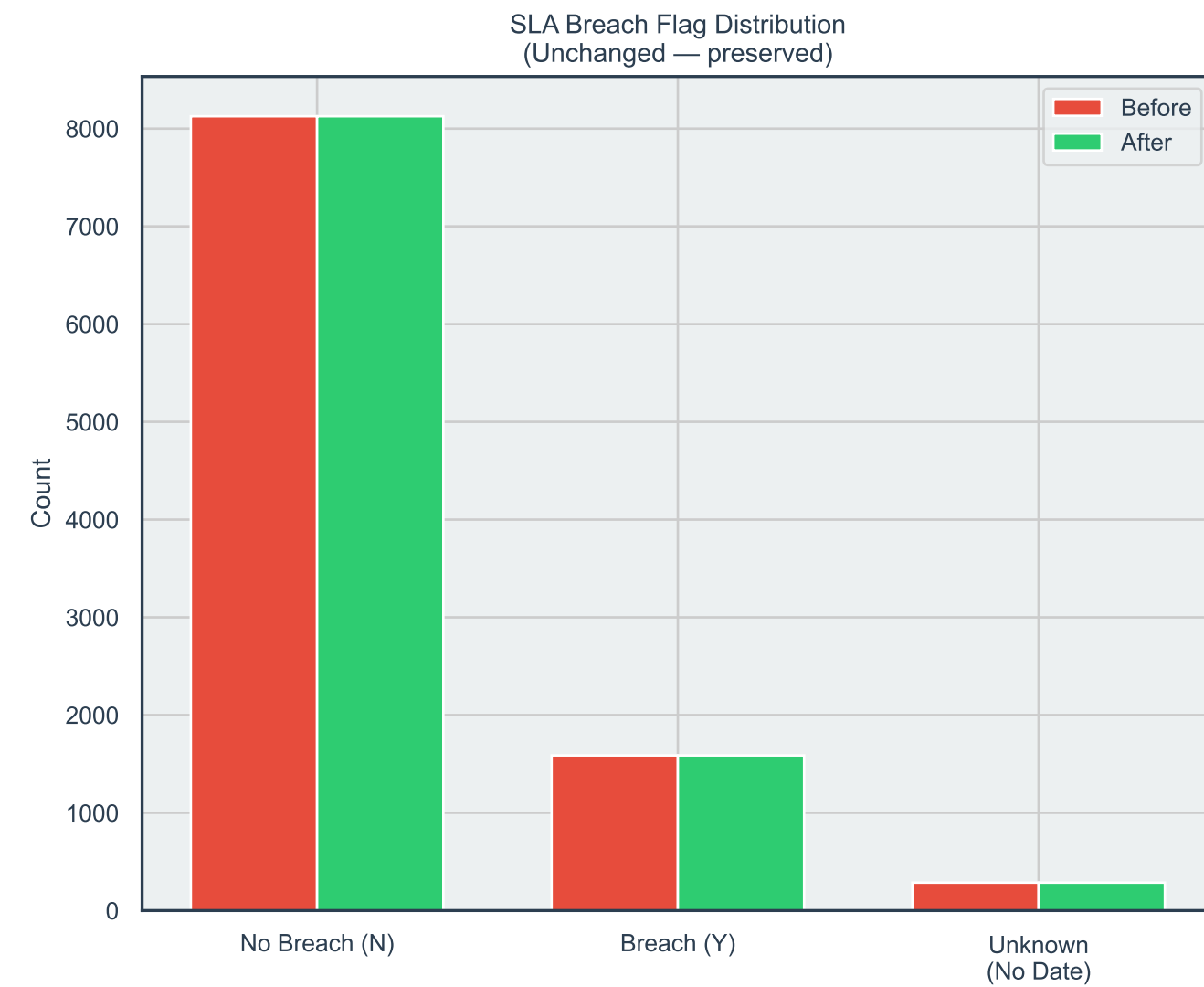


Existing flag: 1,492 | New computed flag: 3,492 | Difference: 2,000 (Auth=Y records not covered by existing feature)

SECTION 6 — SLA & Pipeline Anomalies

Existing SLA features preserved | Pipeline gap supplement added

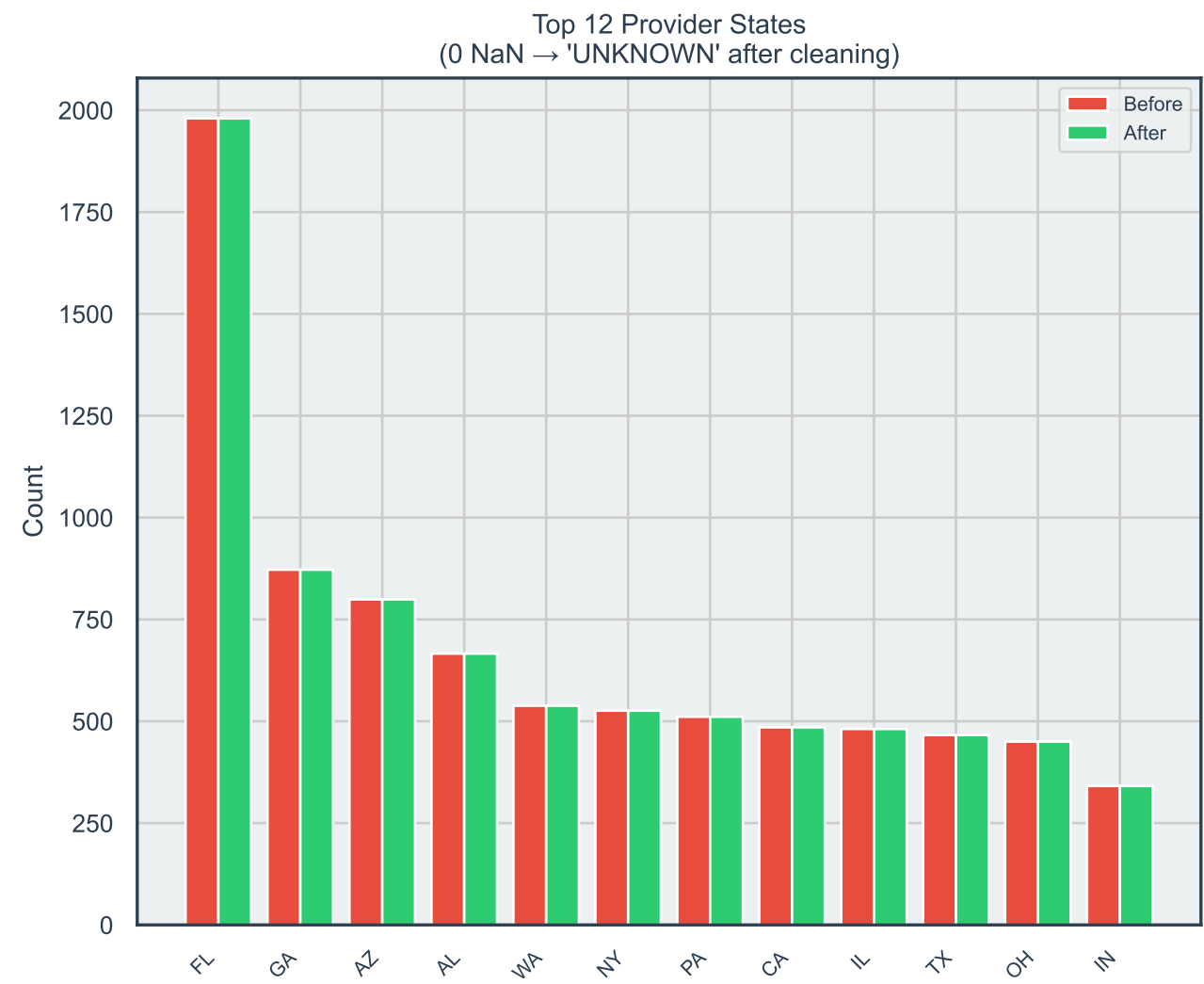
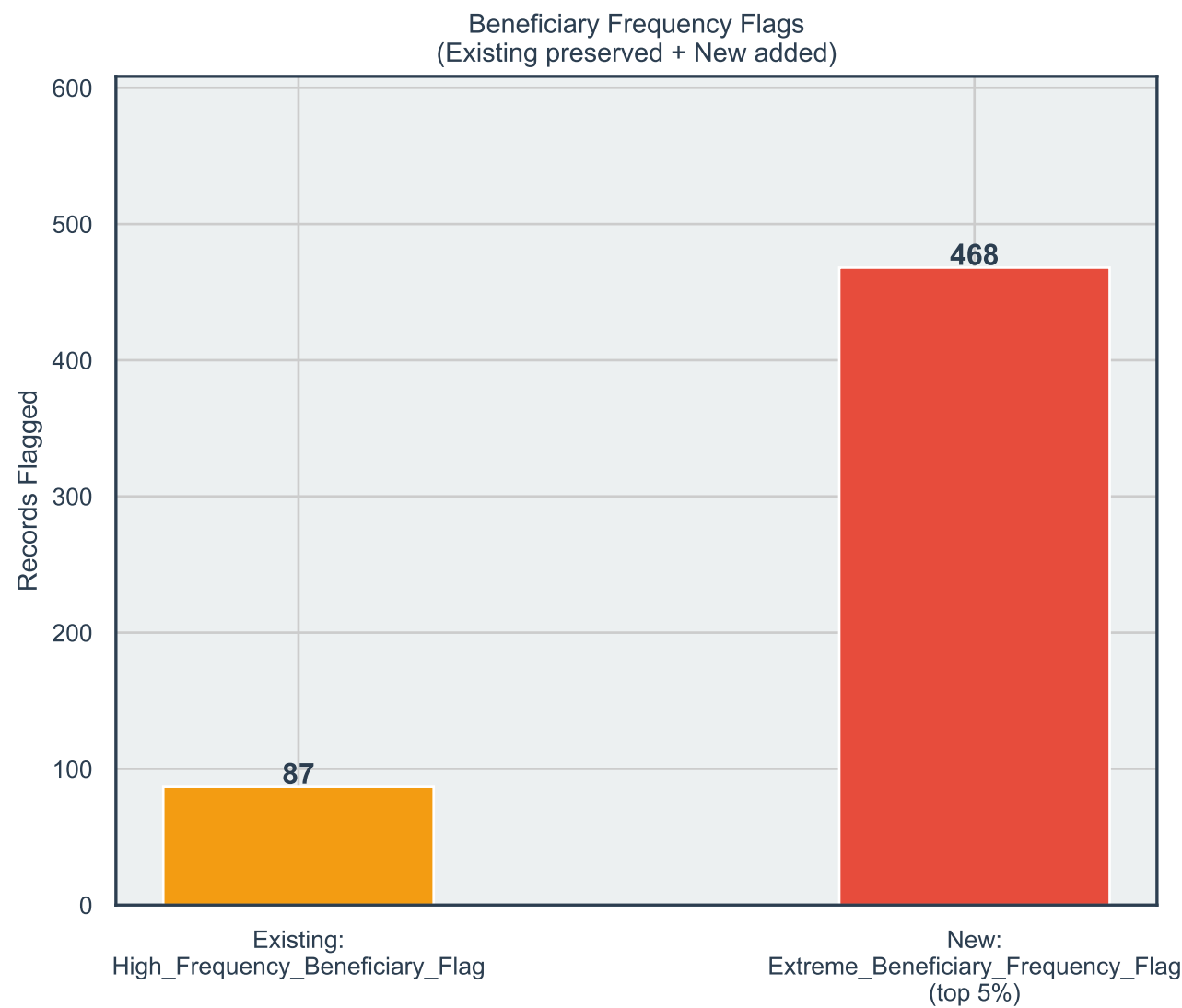
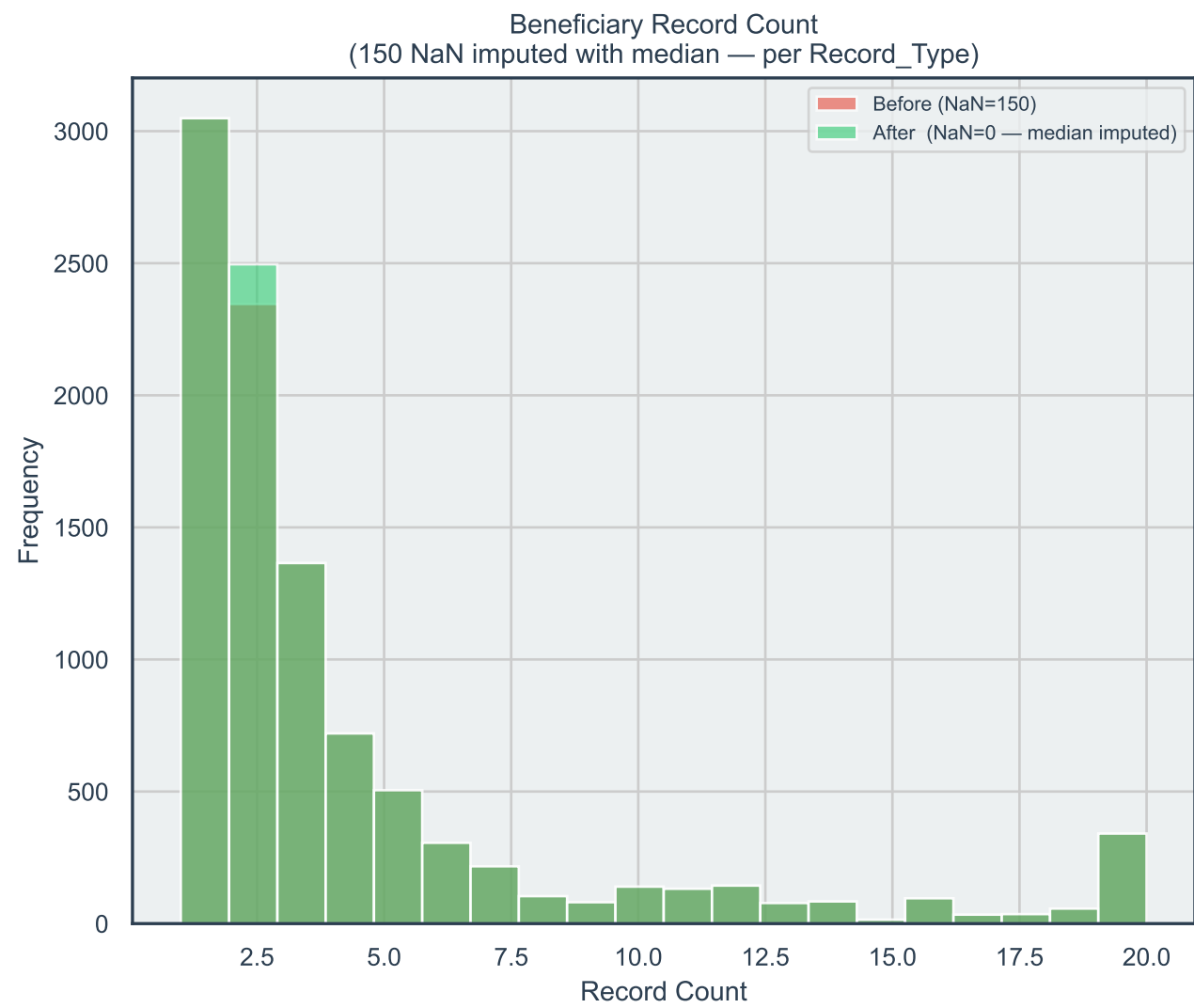
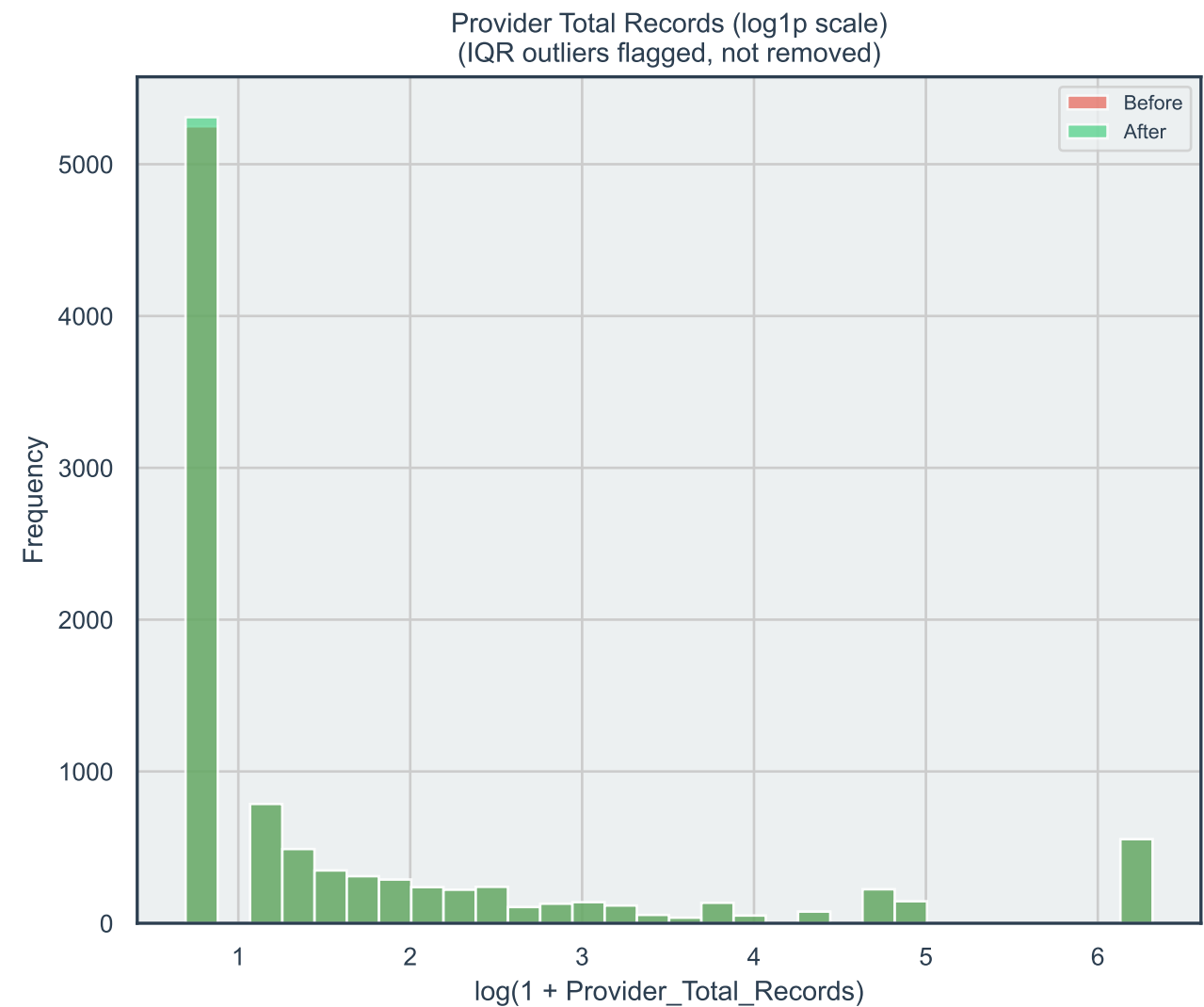
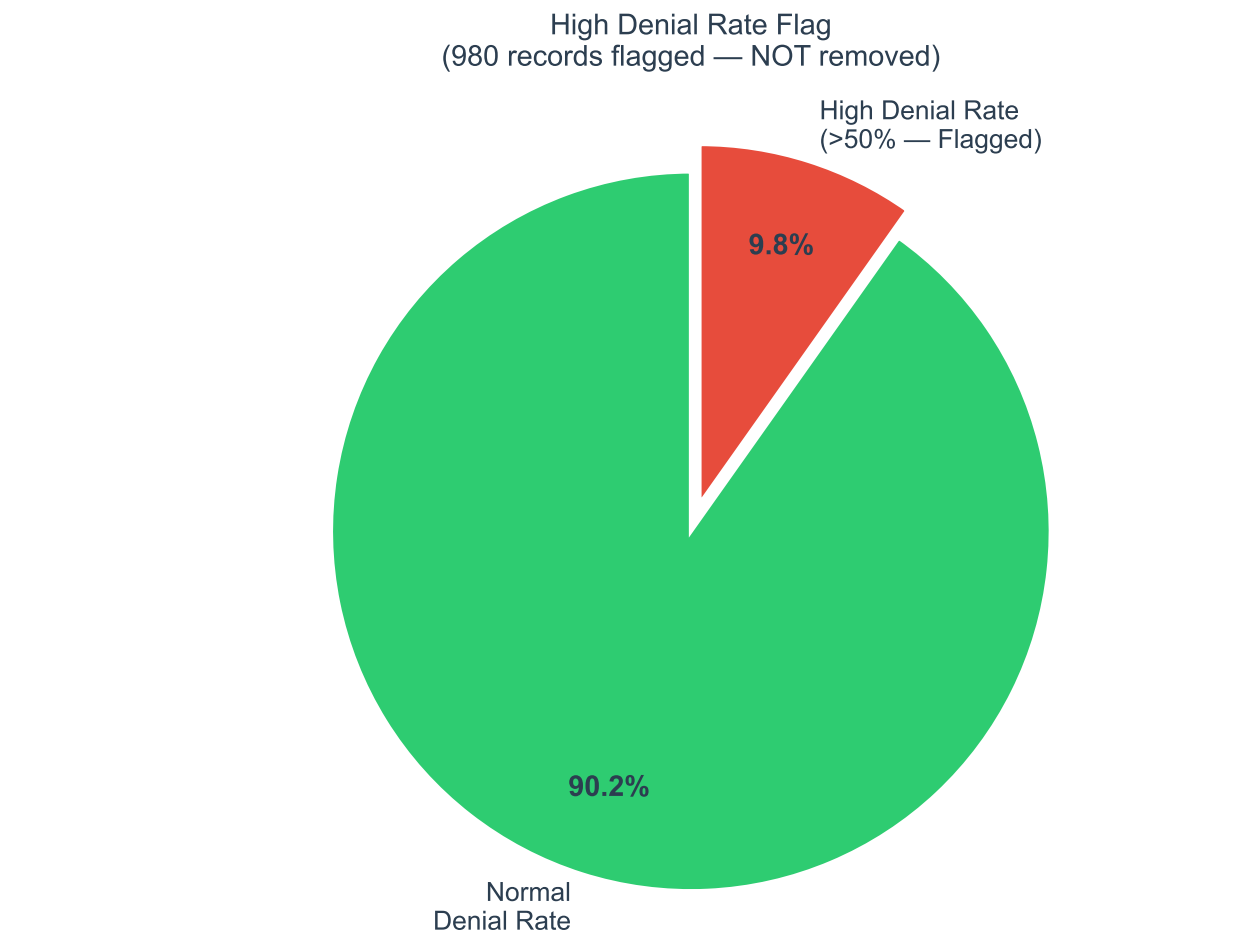
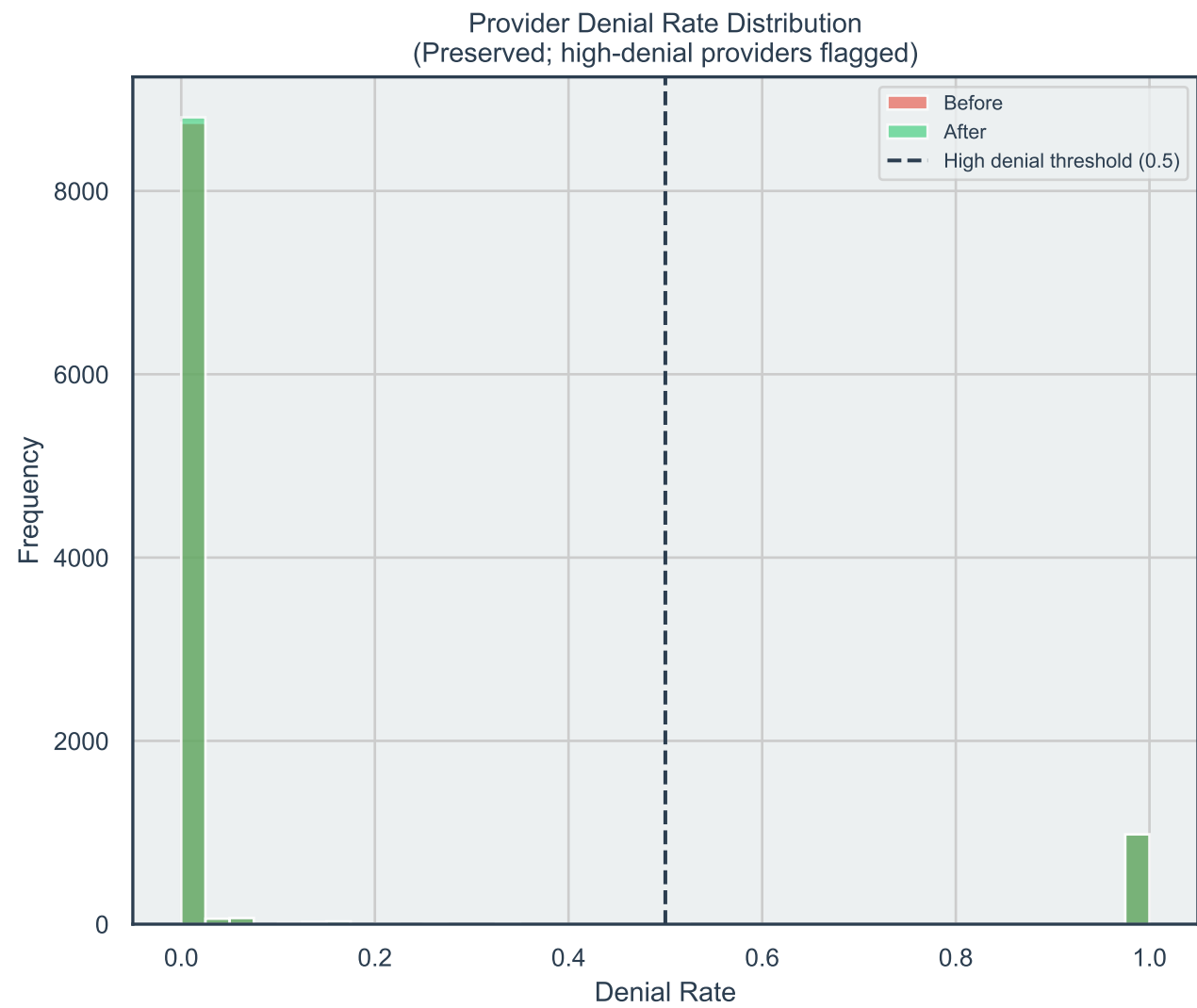
SLA & Pipeline Anomalies — Before vs After



SECTION 7 — Provider & Beneficiary Anomalies

High-denial & high-frequency records flagged, not removed

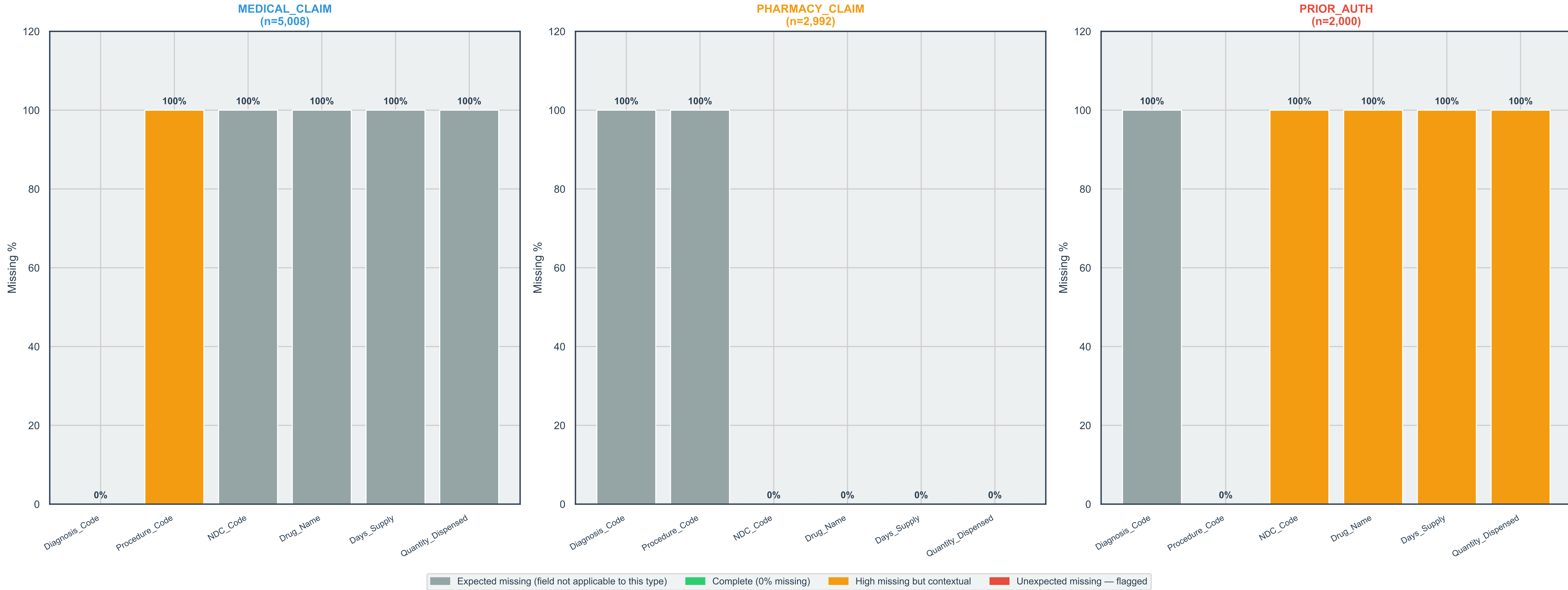
Provider & Beneficiary Anomalies — Before vs After



SECTION 8 — Record-Type-Aware Missingness

Not all NaN is wrong — context-aware handling per MEDICAL/PHARMACY/PRIOR_AUTH

Record-Type-Aware Missingness — Why Not All NaN Is a Problem

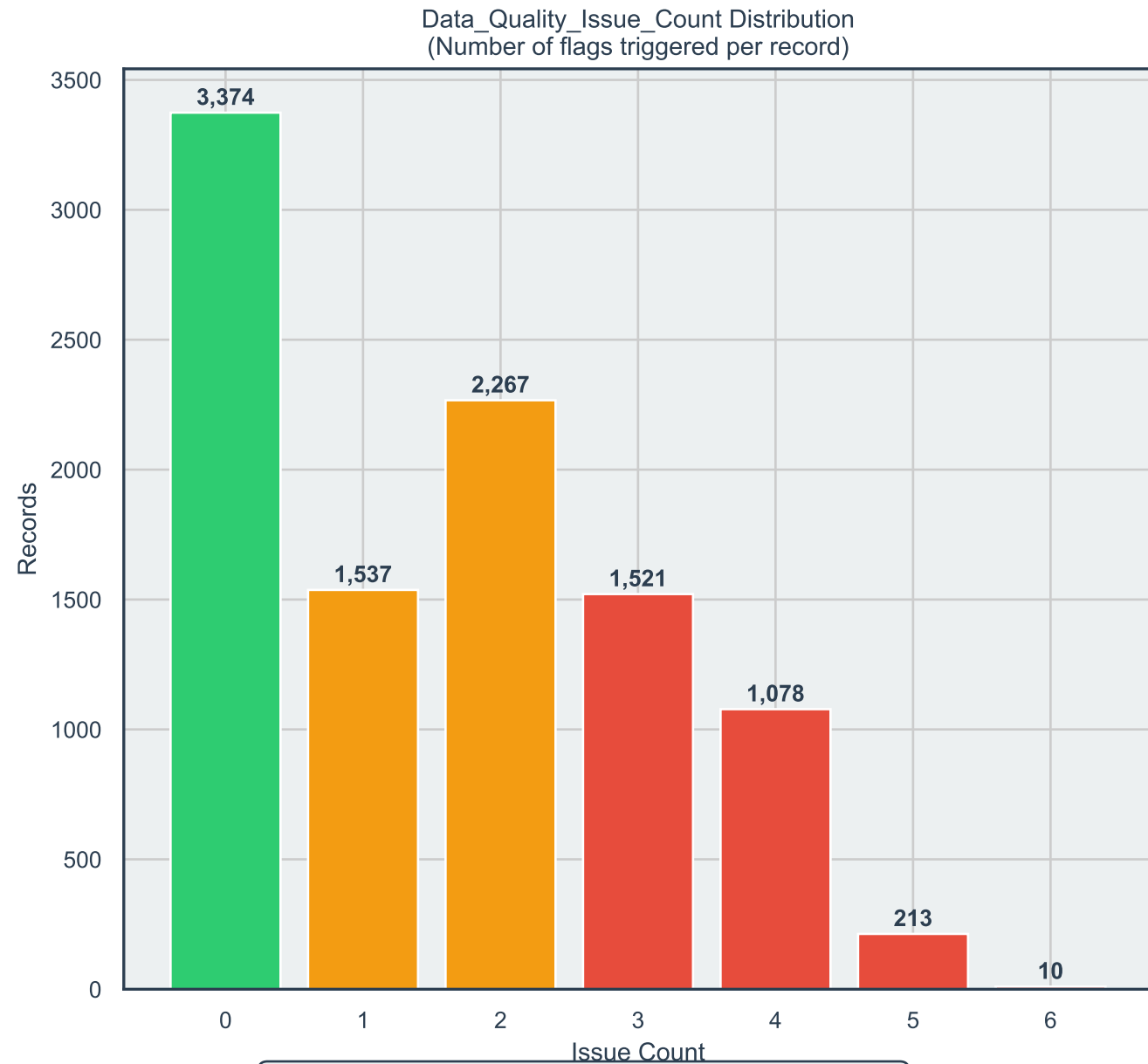
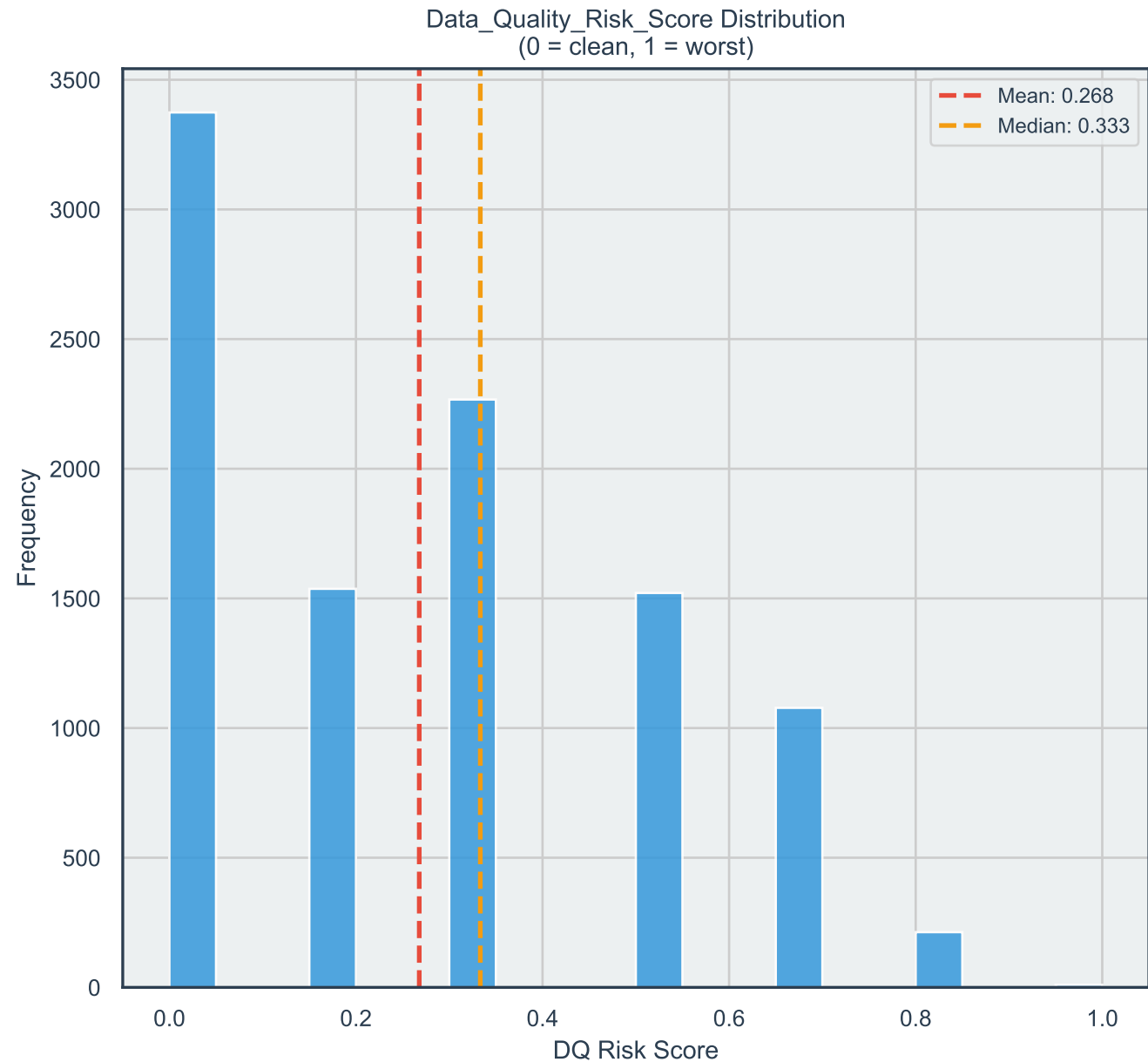


Key insight: NDC_Code/Drug_Name are 100% missing for MEDICAL_CLAIM — that is CORRECT.
Diagnosis_Code is 100% missing for PHARMACY_CLAIM — also CORRECT. These are NOT anomalies.

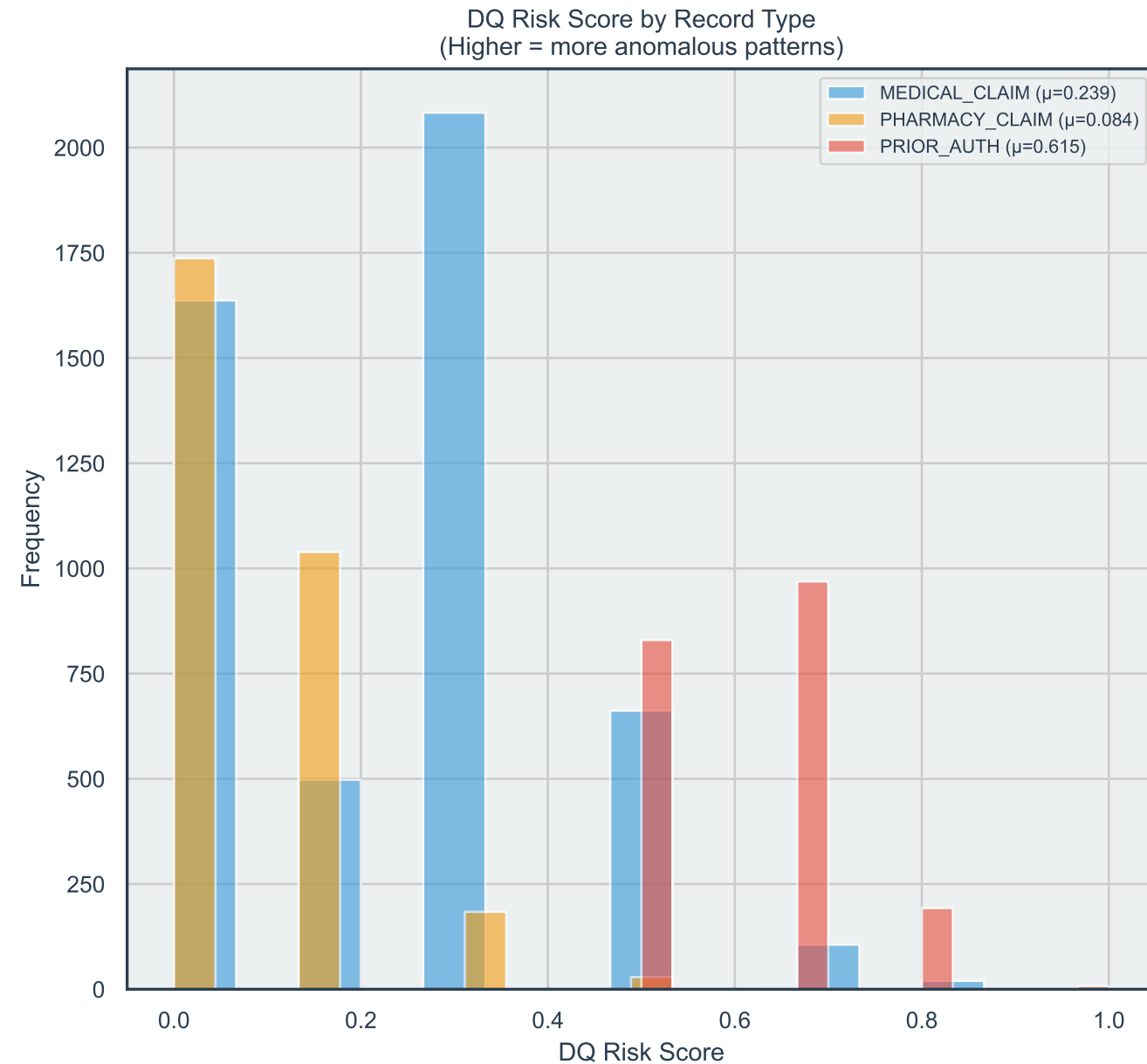
SECTION 9 — Data Quality Score

New composite DQ score — separate from ML anomaly score

Data Quality Risk Score — New Feature Added by Pipeline



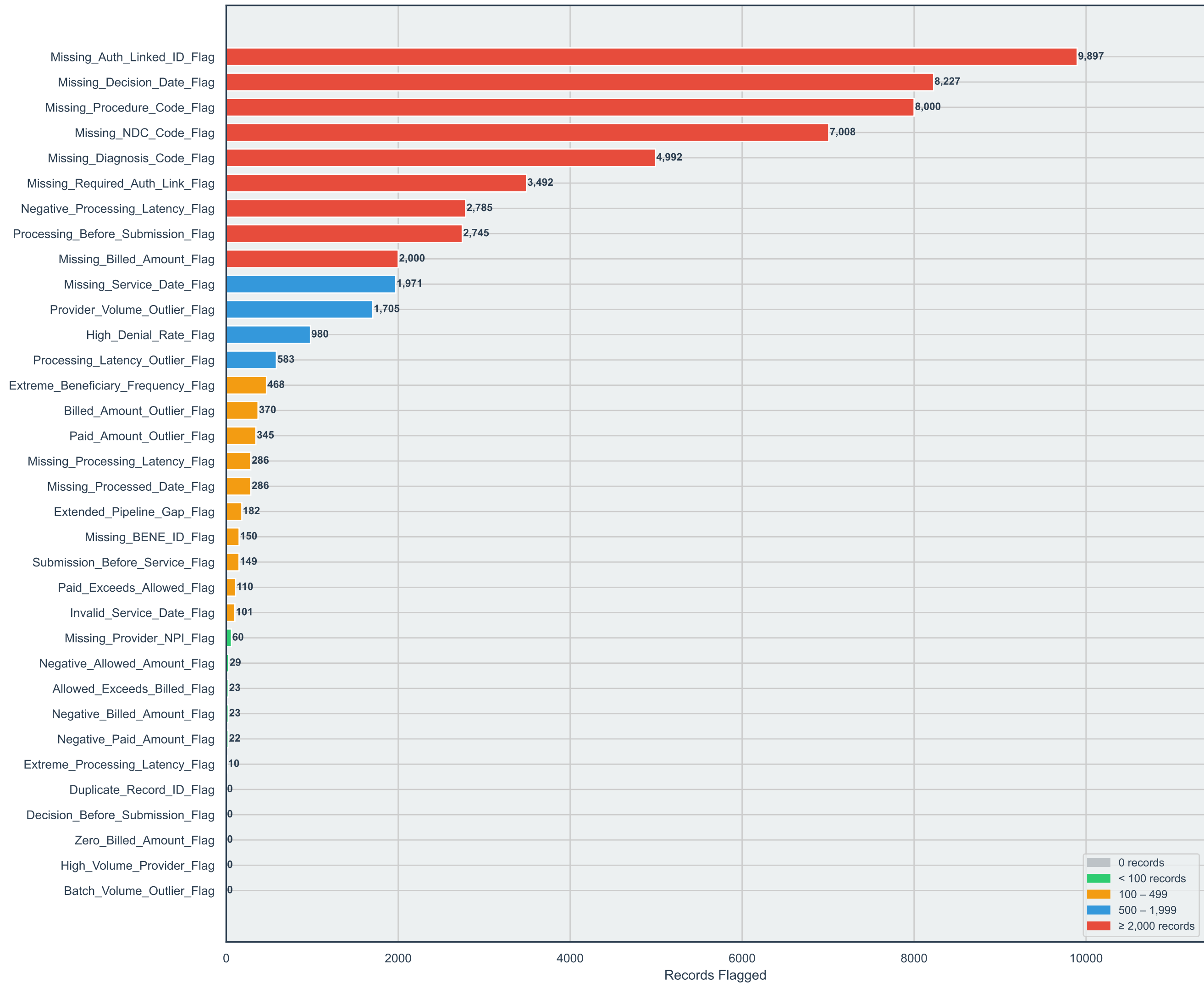
Records with DQ issues: 6,626 / 10,000 (66.3%)
Avg issues per record: 1.61 | Max: 6 flags on a single record
Records with score > 0.5 (High Risk): 1,301



SECTION 10 — All Anomaly Flags

36 new flag columns added — count of records flagged per rule

All New Anomaly / DQ Flag Columns — Record Count per Flag
(Added by Cleaning Pipeline)



SECTION 11 — Cleaning Rules Table

Every rule applied, scope, before state, after state, impact

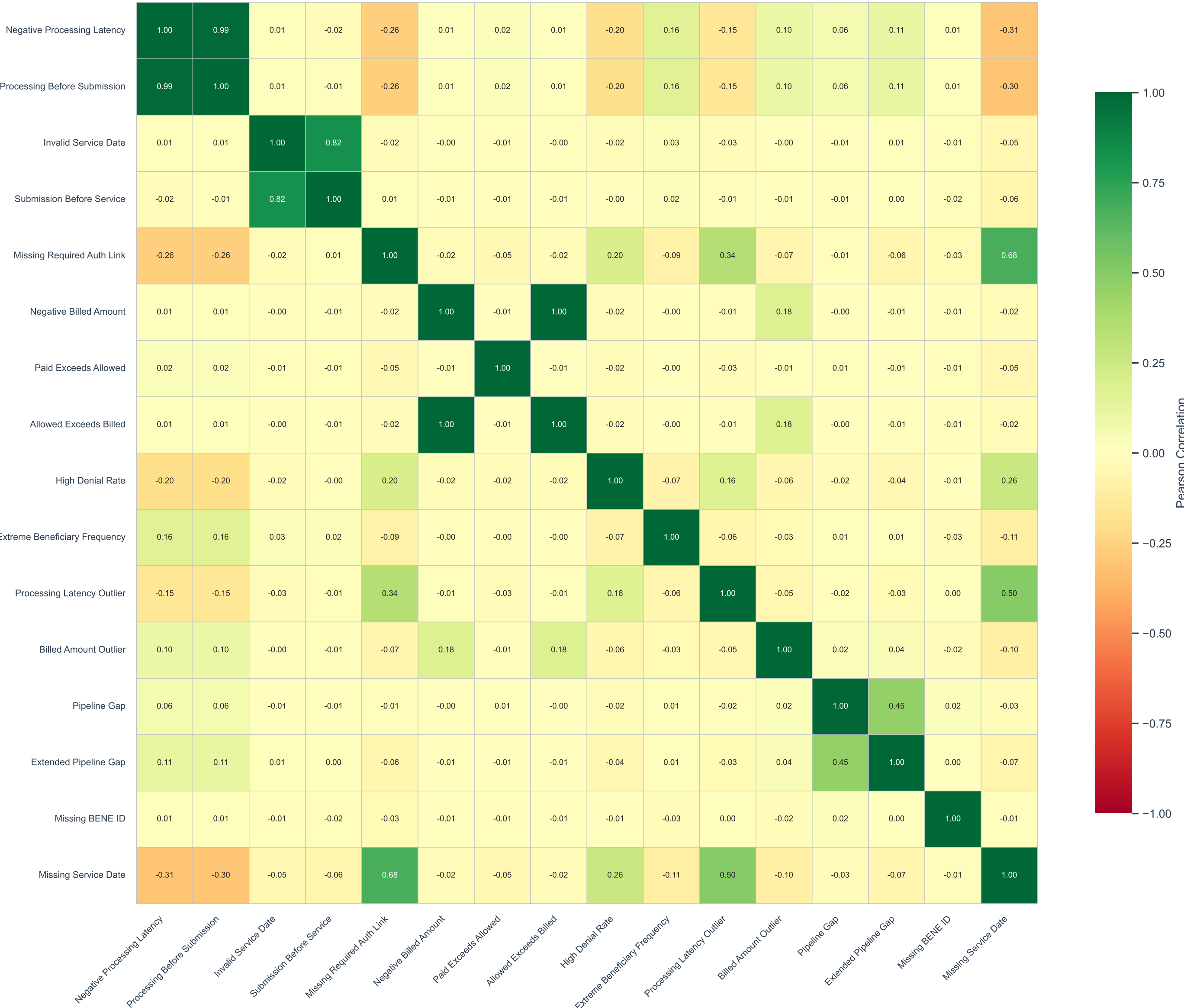
Cleaning Rules Applied — Before vs After Change Summary

Rule Applied	Scope	Before	After	Impact / Records Preserved
Duplicate Detection	All records	0 full duplicates 0 dup Record_IDs	0 removed	No change — dataset clean
Date Parsing	7 date/datetime cols	Stored as raw strings	Converted to datetime errors='coerce'	Enables date arithmetic for anomaly flags
Missing Dates Flagged	Service/Decision/ Processed Date	1,971 – 8,227 missing per col	7 missingness flags created	Records kept; ML can learn from absence
Date Order Anomalies	Service/Submission/ Processed/Decision	101 invalid ranges 2,745 backward process	4 binary flags created	2,846 anomaly records preserved with flags
Neg. Latency Preserved	Processing_Latency_Days	2,785 negative values (pipeline clock issue)	Negative_Processing _Latency_Flag = 1	All 2,785 preserved NOT deleted
Financial Anomalies	Billed/Allowed/Paid Amounts	23 neg Billed 29 neg Allowed 110 Paid > Allowed	5 financial flags created	All 184 records preserved
Auth Link Flagging	Auth_Required_Flag=Y + missing Auth_Linked_ID	3,492 auth required but no link	Missing_Required_Auth _Link_Flag = 1	Critical anomaly preserved for ML
ID Preservation	BENE_ID, Provider_NPI Auth_Linked_ID	150/60/9,897 NaN	Missingness flags NO imputation	IDs never fabricated Flags created instead
Categorical Imputation	Status, Source_System Provider_State	Some NaN values	Filled with 'UNKNOWN'	No data leakage from fabricated IDs
Numerical Imputation	Processing_Latency Provider_Total_Records Days_Since_Prev_Batch	286 / 60 / 18 NaN	Median-imputed per Record_Type	Only non-identifier numerics imputed
IQR Outlier Flagging	Latency, Billed Amt Batch Volume, Provider Vol	Values present but no flags	5 outlier flags created	Original values kept outliers preserved
Provider Flags	Provider_Denial_Rate Provider_Total_Records	No flags	High_Denial_Rate_Flag High_Volume_Provider_Flag	980 high-denial providers flagged
Beneficiary Flags	Beneficiary_Record_Count	High_Frequency flag existed	Extreme_Beneficiary _Frequency_Flag added	Both flags preserved 87/468 records flagged
DQ Score	All 24 flag columns	No composite score	Data_Quality_Issue_Count Data_Quality_Risk_Score	6,626 records have ≥1 DQ issue (66.3%)

SECTION 12 — Flag Correlation Heatmap

How DQ flags co-occur — useful for root-cause analysis

Anomaly Flag Co-occurrence Correlation
(New DQ Flags Added by Pipeline)



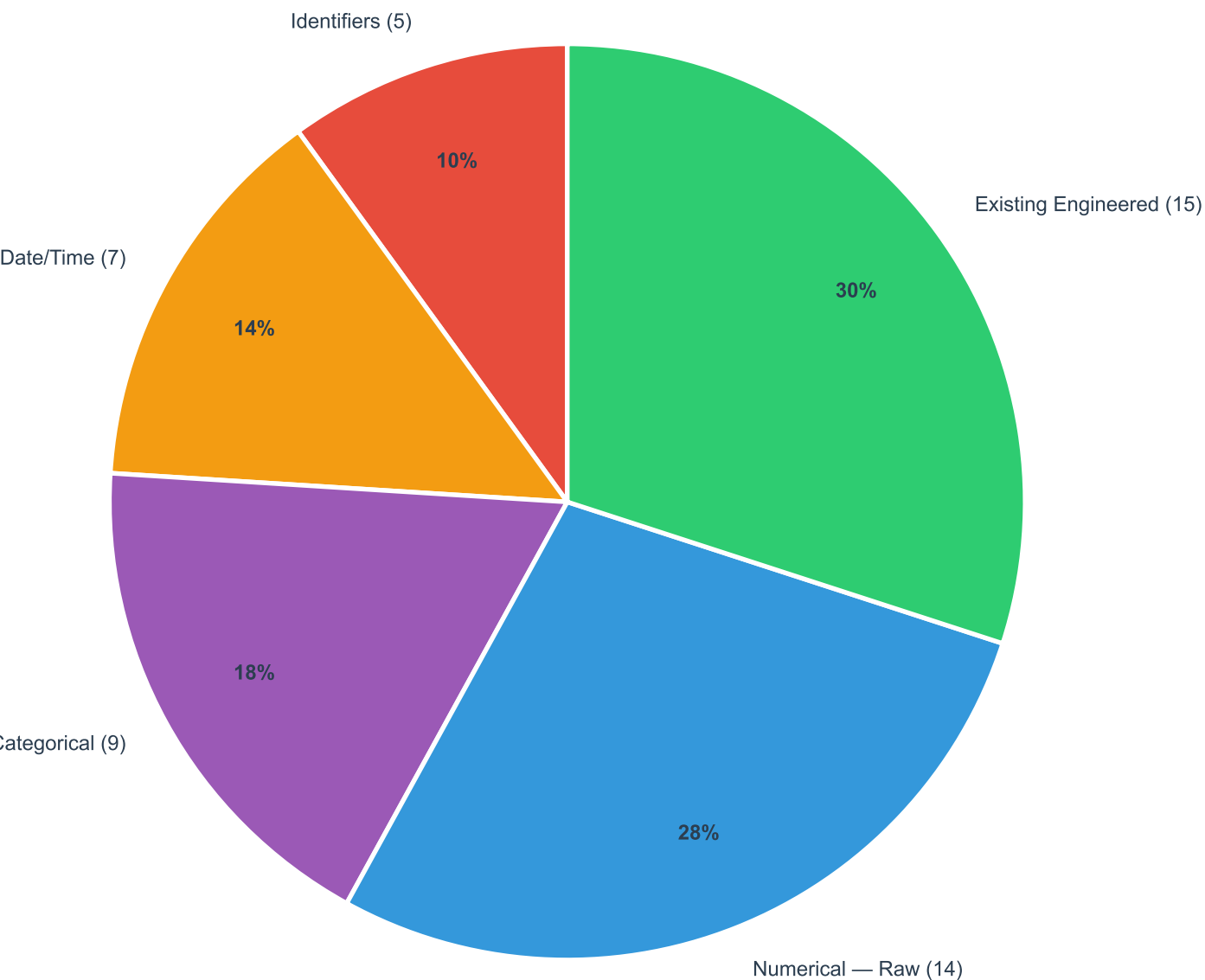
High positive correlation → flags often fire together (same root cause).
Negative_Processing_Latency_Flag ≈ Processing_Before_Submission_Flag (same pipeline clock anomaly).

SECTION 13 — Schema Change Summary

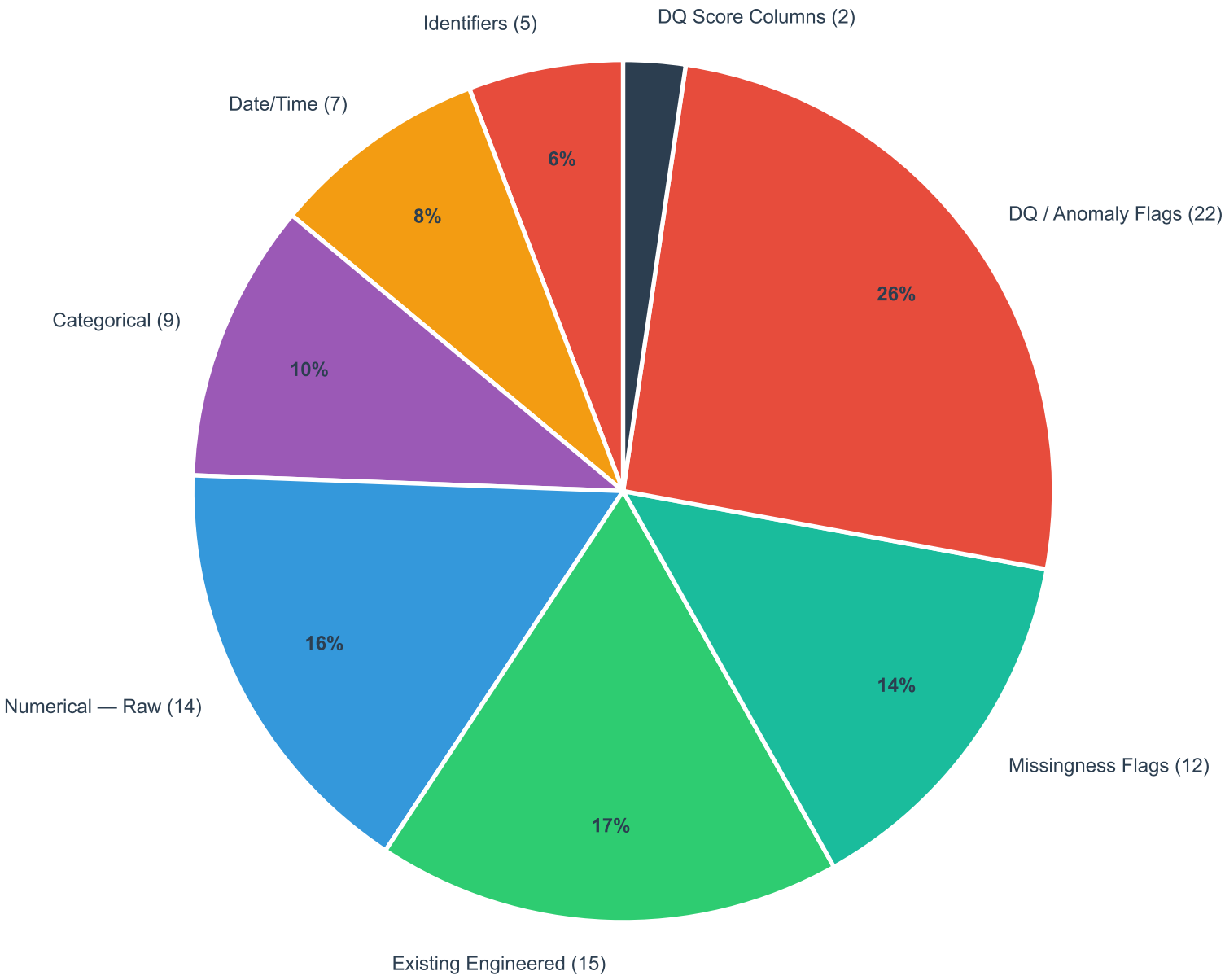
50 → 86 columns | 0 deleted | 36 new DQ/flag columns

Schema Change: Before vs After Cleaning Pipeline

BEFORE Cleaning
50 columns



AFTER Cleaning
86 columns (+36 new)



*Total columns: 50 → 86 (+36 columns added, 0 removed).
All original 50 columns preserved. 36 new flag/score columns added.*